



FOR OFFICIAL USE ONLY

Report No: PADHI00683

INTERNATIONAL DEVELOPMENT ASSOCIATION

PROJECT APPRAISAL DOCUMENT

ON A

PROPOSED GRANT

IN THE AMOUNT OF SDR 73.9 MILLION
(US\$100 MILLION EQUIVALENT)

TO THE

REPUBLIC OF ZAMBIA

FOR A

DIGITAL ZAMBIA ACCELERATION PROJECT (DZAP) (P505094)

AS PART OF PHASE 2 OF THE MULTIPHASE PROGRAMMATIC APPROACH

INCLUSIVE DIGITALIZATION IN EASTERN AND SOUTHERN AFRICA (IDEA) (P512859)

WITH AN OVERALL FINANCING ENVELOPE OF US\$2.48 BILLION EQUIVALENT
APPROVED BY THE BOARD ON JUNE 27, 2024

September 11, 2025

Digital Development Global Practice
Eastern and Southern Africa Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective July 31, 2025)

Currency Unit = Zambian kwacha (ZMW)

ZMW 23.26 = US\$1.00

SDR 0.738056 = US\$1.00

FISCAL YEAR

January 1 - December 31

Regional Vice President: Ndiame Diop

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Division Director: Nathan M. Belete

Practice Manager: Isabel Neto

Task Team Leaders: Tim Kelly, Paul Ndungutse

ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
ASCENT	Accelerating Sustainable and Clean Energy Access Transformation
BoZ	Bank of Zambia
CIRT	Cybersecurity Incident Response Team
COMESA	Common Market for Eastern and Southern Africa
CPF	Country Partnership Framework
CRI	Corporate Results Indicator
DA	Designated Account
DFS	Digital Financial Services
DPC	Data Protection Commission
DPI	Digital Public Infrastructure
DRC	Democratic Republic of Congo
DTC	Digital Transformation Center
DZAP	Digital Zambia Acceleration Project
ECT	Emergency Cash Transfer
ESIA	Environmental and Social Impact Assessment
ESS	Environmental and Social Standards
EU	European Union
FM	Financial Management
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GNI	Gross National Income
GRS	Grievance Redress Service
GSMA	Global System for Mobile communications Association
GWAN	Government Wide Area Network
HEI	Higher Education Institution
ICT	Information and Communication Technology
ID	Identification
ID4D	Identification for Development
IDEA	Inclusive Digitalization in Eastern and Southern Africa
IFC	International Finance Corporation
IPF	Investment Project Financing
IRR	Internal Rate of Return
ISP	Internet Service Provider
IXP	Internet Exchange Point
ITU	International Telecommunication Union
JICA	Japan International Cooperation Agency
KYC	Know Your Customer
LEO	Low-Earth Orbit
M&E	Monitoring and Evaluation
MCTI	Ministry of Commerce, Trade and Industry
MDAs	Ministries, Departments, and Agencies
MIGA	Multilateral Investment Guarantee Agency
MoHAIS	Ministry of Home Affairs and Internal Security
MoTS	Ministry of Technology and Science

MoU	Memorandum of Understanding
MPA	Multiphase Programmatic Approach
NAP	National Adaptation Plan
NDC	Nationally Determined Contribution
NPV	Net Present Value
NRC	National Registration Card
PCE	Private Capital Enabling
PCM	Private Capital Mobilization
PDO	Project Development Objective
PIM	Project Implementation Manual
PIU	Project Implementation Unit
PPA	Project Preparation Advance
PRAMS	Procurement Risk Assessment and Management System
PrDO	Program Development Objective
PSC	Project Steering Committee
PwD	Persons with Disabilities
REC	Regional Economic Community
REN	Research and Education Network
SCT	Social Cash Transfer
SEA/SH	Sexual Exploitation and Abuse/Sexual Harassment
STEP	Systematic Tracking of Exchanges in Procurement
SZI	Smart Zambia Institute
TA	Technical Assistance
TC	Technical Committee
TFSP	Trade Facilitation Support Program
TOR	Terms of Reference
TTL	Task Team Leader
UASF	Universal Access and Service Fund
UCF	Unguaranteed Commercial Financing
UN	United Nations
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UNICEF	United Nations Children's Fund
USAID	United States Agency for International Development
WiFi	Wireless Fidelity
ZAMREN	Zambian Research and Education Network
ZATP II	Zambia Agribusiness and Trade Project II
ZICTA	Zambia Information and Communications Technology Authority
ZRA	Zambia Revenue Authority



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**DATASHEET****BASIC INFORMATION**

Project Beneficiary(ies) Zambia	Operation Name Digital Zambia Acceleration Project (DZAP)		
Operation ID P505094	Financing Instrument Investment Project Financing (IPF)	Environmental and Social Risk Classification Moderate	

Financing & Implementation Modalities

☒ Multiphase Programmatic Approach (MPA)	☐ Contingent Emergency Response Component (CERC)
☐ Series of Projects (SOP)	☐ Fragile State(s)
☐ Performance-Based Conditions (PBCs)	☐ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☐ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☐ Responding to Natural or Man-made Disaster
☐ Alternative Procurement Arrangements (APA)	☐ Hands-on Expanded Implementation Support (HEIS)

Expected Approval Date 11-Sep-2025	Expected Closing Date 31-Oct-2030	Expected Program Closing Date 29-Oct-2032
Bank/IFC Collaboration Yes	Joint Level Complementary or Interdependent project requiring active coordination	

MPA Program Development Objective

Increase access to, and inclusive use of, the internet and digitally enabled services in Eastern and Southern Africa

MPA FINANCING DATA (US\$, Millions)

MPA Program Financing Envelope	2,580.00
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**Proposed Development Objective(s)**

Increase access to, and inclusive use of, the internet and digitally enabled services.

Components

Component Name	Cost (US\$)
1. Affordable broadband	50,000,000.00
2. Interoperable and secure platforms	38,000,000.00
3. Digital Skills and Capabilities	5,000,000.00
4. Program management and capacity building	7,000,000.00

Organizations

Borrower: Republic of Zambia
Implementing Agency: Smart Zambia Institute

MPA FINANCING DETAILS (US\$, Millions)

Board Approved MPA Financing Envelope	2,480.00
MPA Program Financing Envelope:	2,580.00
of which Bank Financing (IBRD):	440.00
of which Bank Financing (IDA):	2,040.00
of which Other Financing sources:	100.00

PROJECT FINANCING DATA (US\$, Millions)**Maximizing Finance for Development**

Is this an MFD-Enabling Project (MFD-EP)? Yes
Is this project Private Capital Enabling (PCE)? Yes

SUMMARY

Total Operation Cost	120.00
Total Financing	120.00



of which IBRD/IDA	100.00
Financing Gap	0.00

DETAILS

World Bank Group Financing

International Development Association (IDA)	100.00
IDA Grant	100.00

Non-World Bank Group Financing

Commercial Financing	20.00
Unguaranteed Commercial Financing	20.00

IDA Resources (US\$, Millions)

	Credit Amount	Grant Amount	SML Amount	Guarantee Amount	Total Amount
National Performance-Based Allocations (PBA)	0.00	100.00	0.00	0.00	100.00
Total	0.00	100.00	0.00	0.00	100.00

Expected Disbursements (US\$, Millions)

WB Fiscal Year	2026	2027	2028	2029	2030	2031
Annual	6.00	15.00	20.00	25.00	26.00	8.00
Cumulative	6.00	21.00	41.00	66.00	92.00	100.00

PRACTICE AREA(S)

Practice Area (Lead)

Digital Development

Contributing Practice Areas

Education; Social Protection & Jobs



CLIMATE

Climate Change and Disaster Screening

Yes, it has been screened and the results are discussed in the Operation Document

SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

Risk Category	Rating
1. Political and Governance	● Moderate
2. Macroeconomic	● Substantial
3. Sector Strategies and Policies	● Substantial
4. Technical Design of Project or Program	● Moderate
5. Institutional Capacity for Implementation and Sustainability	● Substantial
6. Fiduciary	● Substantial
7. Environment and Social	● Moderate
8. Stakeholders	● Moderate
9. Other	● Substantial
10. Overall	● Substantial
Overall MPA Program Risk	● Substantial

POLICY COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No

ENVIRONMENTAL AND SOCIAL

Environmental and Social Standards Relevance Given its Context at the Time of Appraisal



E & S Standards	Relevance
ESS 1: Assessment and Management of Environmental and Social Risks and Impacts	Relevant
ESS 10: Stakeholder Engagement and Information Disclosure	Relevant
ESS 2: Labor and Working Conditions	Relevant
ESS 3: Resource Efficiency and Pollution Prevention and Management	Relevant
ESS 4: Community Health and Safety	Relevant
ESS 5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement	Relevant
ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	Relevant
ESS 7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities	Not Currently Relevant
ESS 8: Cultural Heritage	Relevant
ESS 9: Financial Intermediaries	Not Currently Relevant
NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).	
LEGAL	
Legal Covenants	
Sections and Description	
I. A. 1 a) (a) The Recipient shall establish, not later than thirty (30) days after the Effective Date, and thereafter maintain throughout the implementation of the Project, a Project Steering Committee with composition, functions, and resources satisfactory to the Association, to be responsible for, inter alia, providing overall strategic oversight and governance on the implementation of the Project, including approving the Annual Work Plans and Budgets, and reviewing the progress of the Project.	
I. A 3 a) The Recipient shall establish, not later than thirty (30) days after the Effective Date, and thereafter maintain throughout the period of implementation of the Project, a project implementation unit within the Smart Zambia Institute under the Office of the President, with composition, resources, and terms of reference satisfactory to the Association.	
I.A. 3 c) The Recipient shall recruit/appoint, not later than thirty (30) days after the Effective Date, and thereafter maintain throughout the period of Project implementation, the following specialists, inter alia: a Project coordinator, financial management specialist, procurement specialist, environmental and social specialist, and monitoring and evaluation specialist – all with qualifications, experience and under terms of reference satisfactory to the Association.	
I. A. 4b) The Recipient shall, not later than thirty (30) days after the Effective Date, enter into a memorandum of understanding, in form and substance satisfactory to the Association, with each of the Implementing Partners, outlining the collaboration arrangements, and roles and responsibilities of the Recipient and each of the Implementing Partners, as may be further provided in the PIM.	



Conditions

Type	Citation	Description	Financing Source
Effectiveness	IV. 4.01	The Project Implementation Manual has been prepared and adopted	IBRD/IDA



I. STRATEGIC CONTEXT

1. Zambia will be the fifth operation of the Inclusive Digitalization in Eastern and Southern Africa (IDEA) Multiphase Programmatic Approach (MPA) Project approved by the World Bank's Executive Directors on June 27, 2024, following Republic of Angola, Democratic Republic of Congo (DRC), Republic of Malawi, and the Common Market for Eastern and Southern Africa (COMESA) as the coordinating Regional Economic Community (REC).

A. Country Context

2. **Zambia, a lower middle-income country with about 20 million people, faces multiple human, social, and economic development challenges.** Zambia attained middle-income status in 2011 but faced volatile growth throughout the 2010s, slumping to 1.4 percent in 2019. The COVID-19 pandemic further weakened the economy, leading to a sovereign default in November 2020, and worsening poverty outcomes. Most Zambians work in subsistence farming or informal services, earning low wages due to limited economic diversification and weak links between mining and other sectors. As the economy rebounded from the pandemic, a severe drought in 2024 led to hydroelectric shortages with blackouts exceeding 20 hours per day, disrupting business and digital services. The post-drought recovery has resulted in improved agriculture and mining has accelerated from last year. However persistent low water levels have continued to impact hydroelectric generation, resulting in on-grid supply being available for only 5 to 7 hours per day.

3. **Zambia's young and growing population presents significant opportunities and challenges for job creation and more inclusive growth.** Zambia's population is young and growing rapidly, with 46 percent under 15 years and about 20 percent between 15 and 24 years. The young population means the country is well-placed for a digital leap, but translating this into a demographic dividend will necessitate substantial investments in human capital to create a healthy and skilled labor force, particularly for women, to enable inclusive growth. Although Zambia is one of the most politically stable countries in Africa, weak institutions and limited accountability have hindered service delivery and more inclusive growth. Weak policy coordination and implementation of reforms, as well as limited transparency and accountability, have created space for vested interests to exercise a disproportionate influence in policy-making and allocation of public resources, thereby hindering pro-poor growth. On the positive side, the format of institutions in Zambia (laws, policies, systems, and structures) has shifted to mirror global best practices. The new Government has taken bold steps to stabilize the economy, refocus on private sector-led growth, restore public confidence, and strengthen institutions. Key measures have targeted the debt problem through improved fiscal management, scaling up social safety nets programs, and taking service delivery and resources closer to the people by increasing financing to constituencies while strengthening local governance and accountability.

B. Sectoral and Institutional Context

4. **Digitalization can be a powerful tool for development, transforming entire sectors and delivery of services as well as creating employment.** Higher levels of internet use are associated with significant developmental gains—increases in the number of jobs¹ and a reduction in poverty. Digital can contribute to improved productivity and entrepreneurship. But the current level of use in the Eastern and Southern Africa region is inadequate and uneven. By putting in place a foundational basis for inclusive high-speed broadband networks, secure data hosting, digitally verifiable identification (ID), digital financial services (DFS), and data sharing, countries are establishing the highways of tomorrow that are needed to build inclusive and dynamic digital economies, societies, and governments. The benefits of getting digital public infrastructure (DPI)² right can be enormous: empowered people, improved access to services and economic opportunities

¹ World Bank. 2023. *Digital Africa: Technological Transformation for Jobs*.

² DPI refers to a set of foundational systems and their organizational frameworks (for example, laws and institutions) that enable core functions in today's digital age, including as core elements the ability to verify identities, send and receive money, and exchange data.



for all, more efficient delivery of social assistance, improved data for decision-making and innovation unlocked³. Regulatory harmonization to extend trust across borders can widen the benefits of high-speed broadband, DPI, DFS and citizen-centric e-Government services to facilitate cross-border trade.

5. **Digitalization can also play a critical role in tackling challenges, such as climate change, gender inequality and health emergencies.** Zambia's population is highly exposed to many climate hazards, the most common being floods and droughts (see annex 1). Digital technologies⁴ can help build resilience to the effects of climate change, for instance, by supporting early warning systems for extreme weather events and by developing secure systems for DFS to facilitate electronic cash transfers and remittances.⁵ They also provide a platform for women to access knowledge, resources, and essential services,⁶ especially when supported by appropriate training and strategies. The use of accessible DFS, such as mobile money, has the potential to revolutionize financial services. Digital technologies can also expand access and improve efficiencies and effectiveness of emergency response and aid delivery in fragile contexts as well as to vulnerable communities in remote, rural areas.

6. **Although Zambia has made significant strides on its path to digital transformation, there remain key challenges to inclusivity that the project will address.** These include gaps in internet coverage particularly in rural areas where cell towers have not yet been upgraded to support mobile broadband and fixed line networks are largely absent. Cross-border infrastructure between countries is missing, which is hindering the development of seamless digital trade. There are notable disparities in internet access by women, persons with disabilities (PwD), and older people. Gaps in energy access are a further constraint and efforts to expand internet coverage will be aligned with expansion of energy access.⁷ On the services side, the lack of digitization of government records⁸ and public services means that citizens often have to travel considerable distances and queue to access basic services. Digital skills are also lacking, which prevents the population from getting access to higher paying jobs and the country is not well placed to take advantage of new technologies, such as artificial intelligence (AI) and machine learning. Therefore, the project will tackle three critical challenges of connectivity, digital services and digital skills, which correspond to the three main components of the project.

7. **Although Zambia has a relatively competitive telecommunications market, connectivity is lagging the regional average (Component 1).** With four licensed mobile operators (Airtel, MTN, Zamtel, and Zedmobile), at least six internet service providers (ISPs), and an experienced regulatory authority, Zambia Information and Communications Technology Authority (ZICTA), the country benefits from a favorable environment for further network expansion. The information and communication technologies (ICT) sector has experienced double digital growth and has shown resilience despite the shocks affecting the wider economy. However, the further expansion of usage is constrained by low incomes and relatively high prices, compounded by high excise duties on imported ICT equipment. Internet penetration, at around 21 users for every 100 inhabitants, is below the Eastern and Southern Africa regional average of 28 percent. Although prices for mobile services have become more affordable in recent years, owing to a competitive market, the price of fixed broadband services (5 GB per month) costs more than 12 percent of gross national income (GNI) per capita.⁹ However, the market

³ See World Bank. 2024. Zambia Public Finance Review, Report No. 193530, which highlighted significant data discrepancies from different sources, and governance issues. There is a need to enhance digital administrative data for policy-making.

⁴ Digital technologies include mobile money, emergency communication facilities, internet access, including from schools, government offices, and free wireless fidelity (WiFi) zones, and so on.

⁵ When there are floods and certain areas are inaccessible by road, mobile money platforms, such as Zambia's popular service providers like MTN Mobile Money and Airtel Money, facilitate rapid and efficient delivery of financial assistance during climate-related disasters. This digital financial infrastructure enables emergency funds to reach affected communities swiftly, supporting recovery efforts and reducing the economic impact of disasters. Source: [Digital Frontiers Institute](#).

⁶ The project aims to improve the internet experience for 3.7 million Zambian women through new and enhanced use.

⁷ Of relevance here is the *Mission 300 Initiative* which aims to connect an additional 300 million people in Sub-Saharan Africa by 2030 (see: <https://www.worldbank.org/en/programs/energizing-africa>) and the ASCENT program (P177099). The Digital and Energy teams are closely coordinating the projects in Zambia, notably at the level of digital mapping of demand and infrastructure.

⁸ Paper-based government records are critical to essential service delivery but are also highly vulnerable to climatic events such as floods in Zambia.

⁹ ITU. 2024. *ICT Price Baskets*, data for 2023, at <https://www.itu.int/en/ITU-D/Statistics/Dashboards/Pages/IPB.aspx>.



entry of Starlink, a low-earth orbit (LEO) satellite provider in October 2023¹⁰, is bringing greater price competition and providing new connectivity options in rural areas. The Government's broader goal is to increase internet penetration to 80 percent of the adult population, by 2026 as per the 8th National Development Plan (8NDP), though in 2021 the figure stood at only 21.2 percent, one of the lowest in the region. This will require a significant private sector investment to extend the current 10,000 km of fiber optic cable, and the project aims to add at least a further 2,000 km.

8. **Digital transformation in Zambia will strengthen the economy and create more equitable society by extending the range of digital services available (Component 2).** Digital technologies have a role to play in achieving many of the country's development outcomes. Zambia will increasingly need to rely on digital technologies to ensure that public services, businesses, and individuals are accessible to all. Additionally, these technologies will help withstand current and future hazards and develop a set of resilience measures, to ensure business continuity of government and avoid interruptions in service delivery. A combination of widespread access to broadband, digitally enabled services, and DFS can offer a powerful platform to remove barriers of distance and narrow gender gaps in digital skills as well as internet access and usage. It can also lower costs, improve efficiencies in the delivery of services, create new job opportunities, and enhance human development while also contributing to economic growth and reduced poverty.¹¹

9. **A key element supported by the project is Digital ID.** Zambia is transitioning from its legacy paper-based National Registration Card (NRC) system, introduced before independence in 1964,¹² to a more modern and integrated digital identity ecosystem. The NRC system covered some 94 percent of the adult population by 2021,¹³ and subsequent advances through the Integrated National Registration Information System (INRIS) have registered 15 percent of adults using biometrics. However, the Government is planning to implement a homegrown foundational ID system, based on open standards, and building on the existing investments in order to close the coverage gap of about 85 percent. This new system will prioritize privacy, security, and interoperability, enabling the Government to manage its infrastructure independently and reduce reliance on external vendors.¹⁴ The Ministry of Home Affairs and Internal Security (MoHAIS) is currently collecting minimal demographic data (such as Name, Date of Birth, Address, Gender and biometric attributes -- 10 fingerprints and a photograph); only what is needed to support a unique ID. No sensitive or ethnic data is captured. ID enrollment is free of charge with the first issuance of the ID credential expected to be free. All citizens, marginalized communities, Refugees and Legal residents are eligible for enrollment. Through the ongoing ID4D technical assistance, the government is being supported to develop an operational plan, which includes expanding registration sites via post offices and implementing a public engagement strategy to accelerate registration campaigns.

10. **Digital ID is part of Zambia's broader vision to establish a robust DPI, with digital ID serving as a foundational building block for inclusive service delivery.** Complementary platforms -- such as ZamPay, the Government's eServices portal, ZamSign for e-Signatures, ZamConnect for data integration, ZamPay for interoperable digital payments and ZamConnect for data exchange -- are critical enablers of this vision, facilitating seamless access to public services, enhancing financial inclusion, and driving private sector innovation. ZamPortal, currently offers access to just 300 out of an estimated 1,500 government services. The government plans to provide various authentication options, including via biometrics, smart cards, QR codes, One-Time Passcodes (OTPs) and Paper-Based Credentials with digital features ensuring inclusivity and accessibility for all users. Through integration with various public and private services, this will enable seamless access to healthcare, financial services, and social welfare programs fostering trust and empowering individuals to manage their identity data securely. With technical assistance (TA) from the World Bank's Identification for

¹⁰ TechinAfrica. 2023. <https://www.techinafrica.com/spacexs-starlink-satellite-internet-launches-in-zambia/>

¹¹ GSMA. 2024. "Driving Digitalization of the Economy of Zambia: Leveraging Policy Reforms." The Global System for Mobile communications Association (GSMA) predicts that digitalization of trade in Zambia could add a further 0.39 percent to GDP by 2028 and digitalization of the mining sector would add 0.49 percent.

¹² ID4D Diagnostic of ID Systems in Zambia. 2024.

¹³ [https://databank.worldbank.org/source/identification-for-development-\(id4d\)-data](https://databank.worldbank.org/source/identification-for-development-(id4d)-data)

¹⁴ 2013-14 Zambia Demographic and Health Survey: <https://www.dhsprogram.com/pubs/pdf/SR219/SR219.pdf>



Development (ID4D) Trust Fund,¹⁵ the Government is exploring cost-effective ID credential options, including digitally signed quick response (QR) codes, durable plastic cards, and polycarbonate cards. The World Bank's interventions on Digital ID are guided by the *ID4D Principles*, implemented in close coordination with UNDP, UNHCR and other development partners in Zambia.¹⁶ While significant progress has been made, the Government is still evaluating and will select the most suitable official ID credential to align with its comprehensive DPI strategy.

11. **Enhanced connectivity and increased access to digital services requires a focus on building digital literacy and advanced digital skills (Component 3).** The National Electronic Government Plan 2023-2026¹⁷ emphasizes the need to increase the consumption of e-services by providing both basic and advanced digital skills training through Digital Transformation Centers (DTCs). The government plans to complement basic digital skills with a mobile e-learning application hosting the complete basic literacy curriculum, ensuring continuity and constant learning for citizens. Another priority is a targeted intervention to build a skilled and employment-ready workforce capable of leveraging emerging technologies responsibly. Ongoing discussions are exploring ways to link and complement these digital skills initiatives with other sectors, such as social protection and education, while identifying potential private sector partnerships to effectively scale training efforts.

C. Relevance to Higher Level Objectives

12. **The project is fully aligned with the World Bank's Country Partnership Framework (CPF) for Zambia FY25–29¹⁸ and its supporting Country Economic Memorandum,¹⁹** and digital investments will support all three pillars of more and better jobs, improved human capital, and enhanced climate resilience. The project also responds to the African Union Digital Transformation Strategy, which calls for achieving universal access by 2030. In line with United Nations (UN) Sustainable Development Goal 9, the 2022 Dakar Call to Action, and Agenda 2063, Digital Zambia Acceleration Project (DZAP) will advance the 'Single Digital Market Framework' for East Africa,²⁰ reducing barriers for regional telecom infrastructure and for trade in both digital services and physical goods, through 'no-stop' border facilitated through digital technologies. It will also aid in narrowing gender gaps in internet access, digital ID, and digital skills (see annex 2), in alignment with the World Bank Gender Strategy (FY24–30),²¹ and the baseline study to be undertaken should assist in improving gender-disaggregated data for the ICT sector, which is currently largely absent. The project is aligned with the new World Bank Digital Strategy, approved in July 2025 and the Digital Policy Commitment under IDA-21.

13. **The project is consistent with Zambia's Nationally Determined Contribution (NDC),²² National Adaptation Plan (NAP),²³ and National Policy on Climate Change and will contribute to country's efforts on climate change mitigation and adaptation.** In the latest NDC, Zambia commits to reducing greenhouse gas (GHG) emissions by up to 25 percent by 2030 compared to the 2010 baseline. Zambia's NAP emphasizes the importance of "enhanced social protection programs with improved targeting of beneficiaries," highlighting initiatives such as social cash transfers (SCTs) as key adaptation actions. The project directly supports and enables the Zambian Government's priorities for investments in climate-resilient and low-carbon development as well as in social safety net programs. It does so by integrating climate mitigation and

¹⁵ The World Bank ID4D: <https://id4d.worldbank.org/>

¹⁶ <https://www.idprinciples.org/>

¹⁷ The National Electronic Government Plan (2023-2026) https://www.szi.gov.zm/wp-content/uploads/2023/08/Final-National_e-Government_Plan_-2023-Final-17.08.2023.pdf

¹⁸ World Bank. 2024. Report No CPF0000033. Discussed by the Board on October 7, 2024.

¹⁹ World Bank. 2024. *Zambia Country Economic Memorandum: Unlocking Productivity and Economic Transformation for Better Jobs*.

²⁰ World Bank. 2019. *Single Digital Market for East Africa*. Report 136699. Washington, DC.

²¹ World Bank. 2024. *World Bank Group Gender Strategy 2024-2030: Accelerate Gender Equality to Reduce Poverty on a Livable Planet*.

²² In its revised and updated NDC, Zambia states that it intends to reduce its GHG emissions by 25 percent (at business as usual level of international support prevailing in 2015) and toward 47 percent (with substantial international support) compared to 2010 levels. UNFCC.2022.

https://unfccc.int/sites/default/files/NDC/2022-06/Final%20Zambia_Revised%20and%20Updated_NDC_2021_.pdf

²³ UNFCC. 2023. <https://unfccc.int/sites/default/files/resource/NAP-Zambia-2023.pdf>

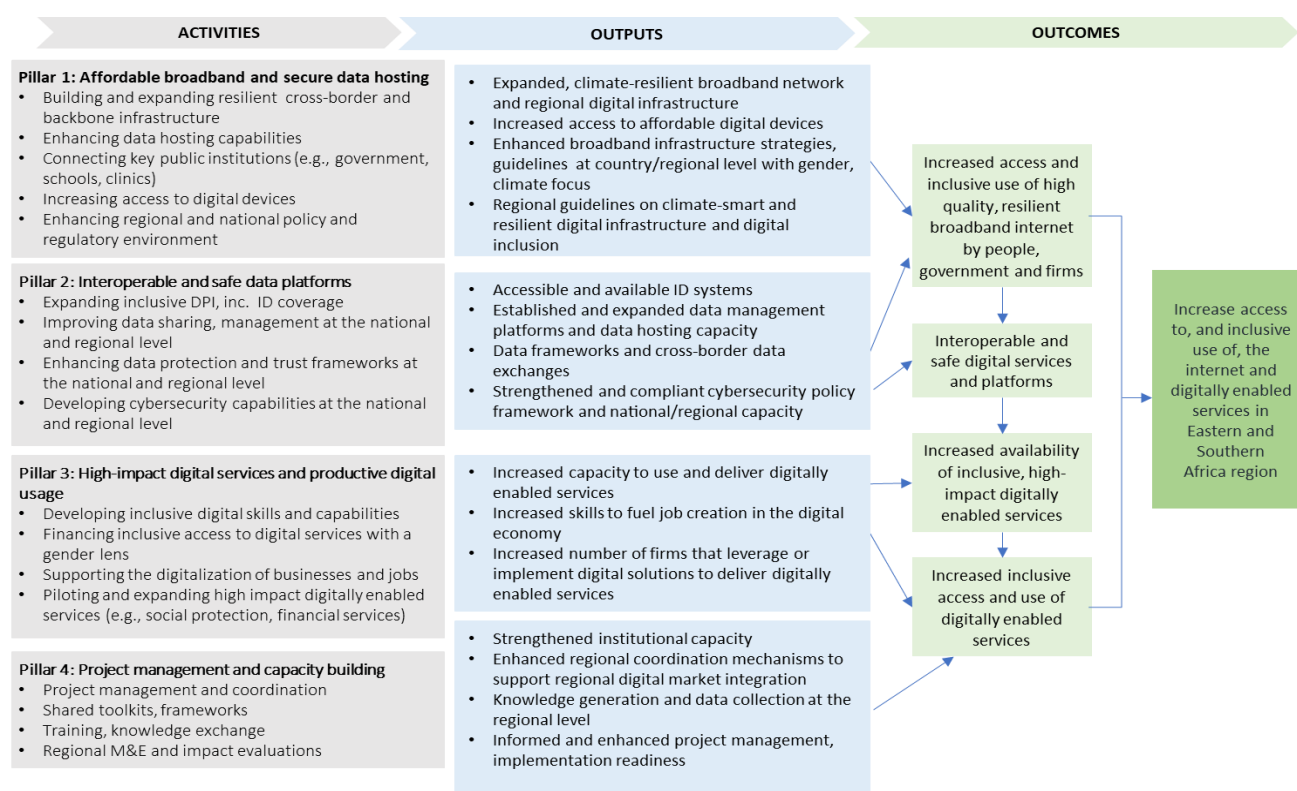


resilience measures in digital connectivity infrastructure and providing access to digital connectivity and hence digitally enabled services such as digital ID and electronic SCTs for communities in climate-vulnerable hotspots.

D. Multiphase Programmatic Approach

14. **Zambia is included in the second phase of countries under the IDEA MPA**, with an appraised amount of US\$100 million in national IDA, plus US\$20 million in unguaranteed commercial financing (UCF). The project design follows a One World Bank approach with close coordination between the IDA, International Finance Corporation (IFC), and Multilateral Investment Guarantee Agency (MIGA) teams. The Government of Zambia, in a letter dated April 17, 2024, requested that Zambia be included in this new regional program and further requested a project preparation advance (PPA) of US\$6 million, which was signed in October 2024.

Figure 1. Theory of Change (IDEA)



15. **There have been no changes to the Program results chain of the IDEA MPA** since its approval by the Board on June 27, 2024, and this is shown in the theory of change in figure 1. The rationale for using an MPA remains valid. The Program Development Objective (PrDO) remains unchanged and is reflected in the Project Development Objective (PDO) for Zambia. The PDO indicators for DZAP are the same as those for the MPA and the targets and timescale have been adapted for DZAP, with a particular focus on adopting the definitions used in the corporate results indicators (CRIs). The planned use of funds in the IDEA MPA is shown on pages 11-12 of the IDEA project appraisal document and summarized below.²⁴

²⁴ <https://documents1.worldbank.org/curated/en/099061024103010133/pdf/BOSIB130fc11f60601aa191c219a13fc1e5.pdf>.



Table 1: Planned use of funds under the IDEA MPA

Phase #	Operation ID	Sequential or Simultaneous	Phase's Proposed DO	IPF or PforR	Estimated IBRD Amount (US\$, millions)	Estimated IDA Amount (US\$, millions)	Estimated Other Amount (US\$, millions)	Estimated Approval Date	Estimated Environmental and Social Risk Rating
1	P502532 COMESA	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Eastern and Southern Africa	IPF	0	10		June 27, 2024	Moderate
2	P180495 Democratic Republic of Congo	Simultaneous	Increase inclusive access and use of the internet, and strengthen the foundations for digitally enabled services in DRC	IPF	0	400	275 (of which 110 co-financing (100 EUR) and 165 UCF)	June 27, 2024	Substantial
3	P180693 Angola	Simultaneous	Accelerate digital inclusion, increase access to digitally enabled services, and unleash digital opportunities for the advancement of Angola's digital economy	IPF	300	0	80 (UCF)	June 27, 2024	Moderate
4	P505095 Malawi	Simultaneous	Increase access to and inclusive use of the internet and improve the government's capacity to deliver digitally enabled services	IPF	0	70	20 (UCF)	June 27, 2024	Substantial
5	P505094 Zambia	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Zambia	IPF	0	100	20 (UCF)	FY26	Moderate
6	P508948 Eswatini	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Eswatini	IPF	65	0		FY26	Moderate
7	Tanzania	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Tanzania	IPF	0	300		FY26	Substantial
8	Botswana	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Botswana	IPF	100	0		FY26	Moderate
9	Malawi (Second Phase) ²⁵	Simultaneous	Increase access to and inclusive use of the internet and improve the government's capacity to deliver digitally enabled services	IPF	0	80	30 (UCF)	FY26	Substantial
10	Lesotho	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Lesotho	IPF	0	40		FY26	Moderate
11	South Sudan	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in South Sudan	IPF	0	150		FY27	Substantial

²⁵ The country is committed to expanding the Program's scope in the near future. Thus, the overall investment in Malawi is split into two operations commensurate to the existing implementation capacity (US\$70 million in the first operation and US\$80 million in the second, thus totaling the US\$150 million appraised amount). This approach is deemed necessary to ensure the successful implementation of the Program in the country.



Phase #	Operation ID	Sequential or Simultaneous	Phase's Proposed DO	IPF or PforR	Estimated IBRD Amount (US\$, millions)	Estimated IDA Amount (US\$, millions)	Estimated Other Amount (US\$, millions)	Estimated Approval Date	Estimated Environmental and Social Risk Rating
12	Madagascar	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Madagascar	IPF	0	200		FY27	Substantial
13	Burundi	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Burundi	IPF	0	125		FY27	Substantial
14	Kenya	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Kenya	IPF	0	200		FY27	Substantial
15	Sao Tome e Principe	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Sao Tome e Principe	IPF	0	40		FY27	Moderate
16	Ethiopia	Simultaneous	Increase access to, and inclusive use of, the internet and digitally enabled services in Ethiopia	IPF	0	300		FY27	Substantial
Total					465	2,015	425		
Estimated Total MPA Financing Envelope					US\$2,480				



II. PROJECT DESCRIPTION

A. Project Development Objective

PDO Statement

16. The PDO for the project is to increase access to, and inclusive use of, the internet and digitally enabled services.

PDO Level Indicators

17. The Results Framework is aligned with the overall IDEA MPA and is set out in Table 2.

Table 2. Results Framework for DZAP

Indicator	Baseline (Dec 2022)	End Target (Oct 2030)		
		New	Inferred	Enhanced
People using broadband internet (new, inferred and enhanced use)	0	3,400,000	2,000,000	2,000,000
• of which female	0	1,700,000	1,000,000	1,000,000
• of which youth	0	571,000	336,000	336,000
• of which, fixed line	0	700,000	n.a	n.a.
People using digitally enabled services (new and enhanced use)	0	2,000,000	n.a	5,000,000
• of which female	0	1,000,000	n.a	2,500,000
• of which youth	0	336,000	n.a	840,000
People with digitally verifiable identification (number)	1,500,000	4,000,000		
• of which female	750,000	2,000,000		
Volume of international data traffic: used international bandwidth in kbit/s per capita (number)	29.8	90.0		

Note: For definitions of new, inferred and enhanced internet users, see M&E plan in section VII.

B. Project Components

18. The expected allocation of US\$100m in IDA grant is shown in table 3, together with the expected contribution of US\$20m from UCF.

Table 3. DZAP Project Components and Funds Allocation

Component	IDA (US\$, millions)	UCF (US\$, millions)	Total (US\$, millions)
1. Affordable broadband	50	18	68
1.1: Extending the backbone and regional network	20	10	3p
1.2: Supplying last mile connectivity	25	8	33
1.3: Strengthening the enabling environment	5	0	5
2. Interoperable and secure platforms	38	2	40
2.1: Developing digital ID services	23	2	25
2.2: Enhancing digital government and services	13	0	13
2.3: Fostering regional trade facilitation	2	0	2
3. Digital skills and capabilities	5	0	5
3.1: Enhancing basic digital literacy	2	0	2



Component	IDA (US\$, millions)	UCF (US\$, millions)	Total (US\$, millions)
3.2: Nurturing employment-ready digital skills	3	0	3
4. Program management and capacity building	7	0	7
Total	100	20	120

Component 1. Affordable broadband (US\$50 million as IDA grant, plus an estimated US\$18 million in UCF)

19. **This component focuses on expanding essential, high-quality, resilient, and affordable broadband infrastructure and services.** Activities under this pillar aim to strengthen institutions and help drive private investment in telecom networks, which are a key foundation for using digital technologies and data productively. It will do this through investments, carried out alongside the private sector, to increase access and improve network speeds and connectivity. This will be done through a One World Bank approach, in close collaboration with IFC and MIGA. Activities are outlined in the following paragraphs.

20. **Subcomponent 1.1: Extending the backbone and regional networks.** The Government plans to extend connectivity to unserved and underserved areas to close the connectivity gap. This can be incentivized by using catalytic funding (matching investments) from project funds, that will be used to fund infrastructure owned and operated by the private sector. The project goal is to add at least 2,000 km of fiber, in the backbone and last mile, and this may be supplemented by funds from the national Universal Access and Service Fund (UASF). It estimated that US\$20 million in project funds in backbone and regional networks could leverage an additional US\$10 million in co-investment in infrastructure from network operators, wholesale infracos including ISPs, mobile network operators, fibercos, satellite operators and other investors, through an efficient allocation of public funds that will attract new market entrants. A further ambition of the Government is to enhance cross-border connectivity by ensuring there are multiple fiber optic border crossings, and resilient loops, with each of Zambia's neighbors. Three of Zambia's neighbors are already IDEA beneficiaries (Angola, DRC, and Malawi). The Government is also interested in establishing further cross-border links through Lake Tanganyika²⁶ and the Lobito corridor and harmonizing telecom policy and regulatory frameworks to align with regional frameworks. This subcomponent will include the following activities:

- (a) Commissioning preparation of a Commercial Transaction Manual (CTM), by a consulting firm selected competitively, to analyze market dynamics, advise on procurement modalities, and guide the mechanics of the competitive procurement process for component 1, by setting out procedures for the PIU to follow, considering likely risks and mitigation measures.
- (b) Undertaking a feasibility study, coordinated with other IDEA countries, to select those backbone, cross-border and cross-lake routes, that will help extend connectivity into noncommercial and underserved areas, including preparation of environmental and social impact assessments (ESIAs). The chosen firm would also monitor infrastructure deployment, but high-level network design will be responsibility of operators bidding. Feasibility studies will be carried out in accordance with World Bank policies
- (c) Conducting a competitive, lot-by-lot bidding process to allocate project funds for infrastructure investment (see Box 1).
- (d) Strengthening government capacity to operate a dedicated government network using dark fiber or managed services from commercial network operators.
- (e) Combined efforts between the Digital and Energy teams of the World Bank and the client to generate digital maps of user demand and infrastructure in Zambia, coordinate site selection, and examine potential for commercialization of fiber held by the electricity utility.

²⁶ See planned links with Burundi, at: <https://extensia.tech/burundi-zambia-agree-underwater-fibre-connection/>



Box 1: Competitive procurement mechanisms to leverage private sector investment

In component 1 of the project, we are proposing to use public funds, from the project, to leverage matching investments from the private sector, to achieve an expanded deployment of infrastructure in the backbone, last-mile and regional networks. These models use the “build, own, operate” principle to achieve sustainability for operations and maintenance of the network. Two particular models*, widely used in other IDA-funded projects, will be employed:

- **“Matching investments”**, whereby project funds are matched by investments from the private sector. Project funds would be allocated through a multi-round reverse auction, administered over an electronic auction platform, in which each route or site to be connected would be one lot. It is anticipated that invitations to tender to the private sector will be issued with indicative IFC and MIGA terms sheets to allow private companies to seek support for their part of the matching investments.
- **“Pre-purchase of bulk internet capacity”** to stimulate demand in the public sector, including schools and HEIs, to encourage the private sector to invest, under an “anchor tenant” model.

The aim is to close the gap between the connectivity needs of rural areas and the financial commitment that network operators are willing to make for financing what might otherwise be an uncommercial investment. The means by which these competitive procurement mechanisms would be deployed will be set out in the CTM to be prepared.

Those routes and sites that benefit from a contribution from project funds would follow World Bank procurement guidelines in the bidding process and would be subject to certain conditions. These would include: (i) the use of open access principles (to allow other operators to use the infrastructure at a reasonable rate); (ii) adherence to climate-resilient standards and use of renewable energy solutions, such as solar power, for power; (iii) the application of World Bank environmental and social standards during construction; (iv) the allocation of a certain capacity (e.g., a number of fiber strands on a fiber optic cable) for the exclusive use of the Government or ZAMREN, for the lifetime of the fiber (i.e. dark fiber). The percentage to be allocated would be proportionate to the relative levels of investment by the Government and the network operator. The use of dark fiber by the Government and ZAMREN will assist with the provision of pre-paid internet capacity under sub-component 1.2; (v) a technologically-neutral bidding process allowing for bidders to propose alternative technologies to fiber in the last mile that meet a minimum required level of functionality (e.g., fixed wireless access, 5G cellular or LEO satellites); and (vi) Service level agreements to be signed between the Government and the winning bidders.

*Note: * These models described in more detail in World Bank/CEPA. 2024. Leveraging Private Sector Investment in Digital Communications Infrastructure in Eastern and Southern Africa. Available at:*

<https://documents1.worldbank.org/curated/en/099135507312434566/pdf/IDU1a0df719e1a15214a9f19be4123f35220101c.pdf>

21. Subcomponent 1.2: Supplying last mile connectivity. This subcomponent aims to extend internet connectivity to a minimum of 500 selected government institutions, including ministries, departments, and agencies (MDAs), including schools, and higher education institutions (HEIs), especially those in climate-vulnerable hotspots.²⁷ If funding becomes available for a second phase, connectivity support could be extended to health facilities, livestock and fisheries camps, refugee settlements and host communities, and so on. Currently just 415 out of the targeted 15,600 national sites have been connected to the Government Wide Area Network (GWAN), leaving a huge infrastructure deficit, and the supply of international internet bandwidth is just 4.5 Gbit/s for the whole of government. Although this segment of the market is less commercially attractive, an investment of US\$25 million in project funds should nevertheless be enough to leverage an additional US\$8 million in infrastructure co-investment. This investment will be used to support the roll out of mobile broadband coverage (4G/5G cellular and/or LEO satellite) and upgrading 2G cell sites to more energy-efficient 4G/5G cellular technologies, with a target of 300 cell sites, as well as installing additional internet exchange points (IXPs) and data

²⁷ Since mid-January 2024, areas in central and southern Zambia have experienced a severe dry spell, devastating nearly half of the country's maize cultivation across 1 million ha and affecting 1 million households in eight provinces, with the most severe impacts in Southern, Central, Eastern, Northwestern, Western, and Lusaka regions.



caches, at the local level, to improve latency.²⁸ As lack of power is a major constraint, site selection will be carefully coordinated with the World Bank's Accelerating Sustainable and Clean Energy Access Transformation Program (ASCENT, P180547) and the International Telecommunication Union (ITU)/United Nations Children's Fund (UNICEF) GIGA initiative,²⁹ for opportunities to extend combined digital and energy access to schools, HEIs, and other selected sites, using renewable energy solutions. Support to schools and HEIs would be coordinated with the Zambian Research and Education Network (ZAMREN) and aligned with eGovernment activities to be supported under Sub-component 2.2.

22. This subcomponent will include the following activities:

- (a) Selecting sites, based on Government priorities, and preparing ESIA's, using the same consultancy as for Subcomponent 1.1, with active participation by the respective line ministries and in close coordination with the ASCENT and GIGA teams, with coordinated development of a digital map of sites and infrastructure.
- (b) Pre-purchase of bulk internet capacity, with coordination at the regional level with other IDEA beneficiaries, under long-term supply agreements (typically 5–10 years), to support the future needs of government users, including in government-supported public free WiFi hotspots, and especially those activities targeted under different project activities (notably under Subcomponents 2.2 and 3.1).
- (c) Conducting a competitive, lot-by-lot bidding process (combined with Subcomponent 1.1c above) to allocate project funds to those network operators bidding (see Box 1).
- (d) Supporting ZAMREN, including covering its membership fees to join the European Union (EU) Africa Connect 4 program,³⁰ to leverage additional funds and assistance for open educational resources.

23. **Subcomponent 1.3: Strengthening the enabling environment.** This subcomponent aims to improve the overall policy and legislative framework governing digital infrastructure, promoting market competition, affordability, and quality of service. The Government is keen to explore options for broadband market development, optimization of the UASF, and key incentives to attract investments in expansion of networks and technologies such as 5G. To minimize the environmental impact of ICT infrastructure and ensure energy efficiency, the Government aims to promote green digital technologies, such as green data centers, satellite connectivity and responsible e-Waste management, as stipulated in the National e-Government plan. The Government is currently reviewing some policies including an AI policy, a 'dig-once' policy, an infrastructure sharing policy, and reviewing the ICT Act, and these will be supported under the project. Additionally, the project will support secure online transactions through electronic signatures, improving efficiency in public services and digital banking.³¹ Additional support will be provided to strengthen trust services by operationalizing the Data Protection Commission (DPC), established in 2021 and currently hosted by the Ministry of Technology and Science (MoTS), and the Cybersecurity Incident Response Team (CIRT).

24. This subcomponent will include the following activities:

- (a) Assessing the current policy and regulatory landscape and institutional arrangements in the ICT sector to inform downstream investments in the project, and appropriate policy reforms.
- (b) Supporting implementation of policy and regulatory reforms, as per the recommendations of such assessment, including in areas such as operationalization of the UASF, and adoption of dig-once and infrastructure-sharing policies.

²⁸ World Bank. 2024. *Green Digital Transformation: How to Sustainably Close the Digital Divide and Harness Digital Tools for Climate Action*. Climate Change and Development Series.

²⁹ <https://www.unicef.org/innovation/giga>

³⁰ <https://africaconnect4.net/>. ZAMREN currently manages regional research and education network (REN) traffic for DRC, Malawi, and Zimbabwe on behalf of EU AfricaConnect 3.

³¹ Digital public infrastructure (DPI) policy note on e-signatures. <https://openknowledge.worldbank.org/entities/publication/d56f94c3-c1c8-4b17-b479-fd68f9551b1c>



- (c) Providing TA, practical support and ICT equipment for operationalization of the DPC. This will include (i) developing an implementation strategy including a governance structure; (ii) establishing standard operating procedures and sector-specific guidelines; (iii) support for hosting stakeholder consultation workshops; (iv) building capacity for data controllers and licensing of data auditors and data processors including knowledge exchange to learn from advanced data protection offices; (v) acquiring a digital platform and related ICT equipment to manage and monitor data protection functions, (vi) Support to DPC in conducting user awareness campaigns and consumer education.
- (d) Providing TA for enhancement of cybersecurity incident response capabilities for Government. For government cybersecurity, this will involve: (i) TA for establishing a government Security Operations Center mandated for operational monitoring and protection of government network and systems; (ii) enhancing the Cybersecurity Incident Response Team (CIRT), enabling it to perform strategic coordination, information sharing, and national incident response; and (iii) strengthening cybersecurity capabilities and capacities. Additionally, for overall national cybersecurity capacity, the project will support (iv) fostering knowledge exchange and collaboration; and (v) conducting cyber awareness campaigns and consumer education.
- (e) Developing a policy and operationalizing and updating the National Strategy for AI and regulations for emerging technologies and building capabilities in emerging technologies such as big data, blockchain, machine learning, and robotics, including support for capacity building and developing a local language library.
- (f) Developing e-waste regulations and guidelines and supporting establishment of e-waste disposal/recycling facilities, including support for awareness campaigns and consumer education.³²
- (g) Support for a baseline statistical study of ICT usage, in collaboration with ZICTA and ZAMstats, to take place in mid- 2025, which will inform the project results framework.

Component 2. Interoperable and secure platforms (US\$38 million as an IDA grant, plus US\$2 million in UCF)

25. This component will focus on deploying interoperable DPI, notably digital ID, to enhance efficiencies in both public and private sectors, strengthen institutions, and develop cross-cutting enablers and digital safeguards needed to promote adoption of digital technologies and digitally enabled services at the national level and across borders. Activities under this component are outlined below.

26. **Subcomponent 2.1: Developing digital ID services.** The Ministry of Home Affairs and Internal Security (MoHAIS), through its Department of National Registration, Passport, and Citizenship (DNRPC), is legally mandated to issue NRCs and collect biometric data. As part of its modernization efforts, Zambia is transitioning from a legacy paper-based ID system to a more advanced, secure, and inclusive digital identification system. Around 11 million out of an estimated 20 million legacy ID records have already been digitized, but there is a need to further modernize the entire ID ecosystem in the country and advance toward a national digital ID system. With technical support from the World Bank's ID4D team, the Government has chosen to develop a homegrown digital ID enrollment system based on open standards. This approach enhances privacy and security, reduces dependency on vendors, and allows the Government to maintain control over its infrastructure, ensuring a more flexible and cost-effective system. Biometric data has been collected for around 1.5 million people so far and the project targets increasing this to at least 4 million.

27. **Zambia's transition to a fully paperless system requires an upgraded digital ID framework** capable of supporting digitally verifiable credentials and authentication processes. A modernized system with enhanced biometric capabilities will ensure robust, secure, and reliable identification for all citizens, enabling seamless access to digital services and transactions. Integrating the digital ID with other key registries, such as the social registry and farmer registry, is critical to

³² The development of e-waste regulations, guidelines, and disposal facilities will primarily integrate and focus on reuse, repair, and recycle, and will be coordinated with the Zambia Environmental Management Agency (ZEMA) and ZICTA. ITU is also providing support.



improving the planning and delivery of government programs, streamlining identification and verification services for beneficiaries of programs such as social protection and agricultural subsidies, including e-vouchers ensuring that services reach their intended beneficiaries efficiently. This shift will not only enhance service delivery but also reduce fraud and ensure efficient, direct delivery of benefits such as cash transfers and subsidies. By simplifying know-your-customer (KYC) processes, digital ID will significantly enhance access to financial services through empowering marginalized groups, such as women and rural communities to open bank accounts, access credit, and utilize digital payment systems to participate more in the digital economy.

28. **The 2024 ID4D diagnostic report outlines key legal reforms needed to strengthen Zambia's Digital Identity System.** Proposed changes include updating the National Registration Act to allow additional biometric data collection, such as iris scans, and simplifying documentation requirements for marginalized groups. The Government also plans clarifying dual citizenship and streamlining naturalization processes under the Citizenship Act, revising the Birth and Death Registration Act to include non-medical facility registrations, and harmonizing the Marriage Act for equal recognition of all types of marriage. These reforms will be supported through TA to ensure effective implementation and capacity building for stakeholders involved. This will provide expertise and resources to guide the revision of legal frameworks, develop comprehensive operational strategies, and establish mechanisms for enforcing the updated laws.

29. This **digital ID subcomponent** will include the following activities:

- (a) Competitively procuring solar-powered biometric ID registration kits and PKI as a service.
- (b) Competitively procuring a mobile based digital authentication and an upgraded e-signatures solution.
- (c) Accelerating the rollout of ID registration with the addition of at least 4 million people, using an enrollment system based on open standards. The project aims to increase the number of women and girls with IDs by implementing targeted awareness campaigns, supporting enrollment initiatives, simplifying application processes with agent facilitation, and ensuring well-staffed and accessible enrollment centers.
- (d) Enhancing capacity building and knowledge exchange to ensure application of best practices in digital ID.
- (e) TA to implement key legal reforms.

30. **Subcomponent 2.2: Enhancing digital government and services.** This subcomponent will support the upgrade of ZamPay from its current focus on person-to-government (P2G) payments to an integrated government-to-people platform, forming part of the national DPI. This will strengthen the platform's scalability, allowing it to support a range of citizen-facing programs, based on user demand, including social protection programs offering a scalable and resilient payment system, especially during periods of crisis and shock. By integrating the digital ID system with social registries and payment platforms, the project will enable efficient targeting and disbursement of social protection benefits, such as cash transfers and subsidies, directly to beneficiaries. This reduces delays and leakages, ensuring timely support for vulnerable groups. Additionally, the streamlined KYC processes enabled by the digital ID will expand access to DFS by empowering marginalized groups, such as women and rural communities, to open bank accounts, access credit, and utilize digital payment systems. The improved ID framework will also simplify authentication for e-services, enabling citizens to access services such as healthcare, financial services, education more conveniently and securely. By fostering inclusivity, this subcomponent ensures that the digital ID becomes a gateway to greater social and economic participation for all citizens. ZamSign will be enhanced and upgraded to provide a secure, reliable and robust e-Signatures solution, facilitating seamless digital signing and authentication for a wide range of transactions and services. Additionally, the project will onboard key government agencies including local authorities, agricultural institutions, and the Zambia Revenue Authority (ZRA) onto ZamConnect to enable secure data sharing between government entities. The Government is interested in leveraging cloud storage services as more government services are digitized. This will require a data classification exercise



to distinguish between those few services that need localized data storage and the majority of services that do not handle sensitive data and can therefore be hosted on the cloud.

31. **This subcomponent will support digitization and business process reengineering efforts in key sectors where digital transformation is expected to deliver significant impact.** An initial assessment of the existing digital landscape is being conducted under the PPA to identify priority government services to digitize. Additional support will be provided to digitize these services and develop transaction-based portals, facilitating seamless user interaction with the ZamPortal. Priority will be given to projects within the World Bank's portfolio where digital tools serve as critical enablers, ensuring close collaboration between the Government and the World Bank teams, particularly in social protection and DFS, and supporting local government. Other sectors, such as health, may also be targeted at a later stage if additional IDA funding becomes available. This subcomponent will include the following activities:

- (a) Operationalizing the Government Enterprise Architecture, currently under development.
- (b) Upgrading the ZamPortal including conducting a business process reengineering to streamline workflows, improve the efficiency, and accelerate adoption of government e-Services, and ensure they are accessible to persons with disabilities.
- (c) Upgrading ZamPay into a scalable platform, enabling secure payments, to and from Government.
- (d) Digitizing and onboarding selected citizen-centric government services onto the ZamPortal, based on user demand.
- (e) Digitizing government legacy records including digital ID, and births and death registration, and reengineering business processes to ensure improved service delivery.
- (f) Developing a government cloud strategy to inform implementation and adopting secure cloud services.³³
- (g) Developing a data classification and governance framework with guidelines to migrate legacy records to digital format, including migration to cloud-based services.
- (h) Onboarding key government agencies to the government data exchange platform ZamConnect to enable secure data sharing between government entities.
- (i) Providing TA to Smart Zambia Institute (SZI)³⁴ on emerging technologies and conducting digital and information systems audits to strengthen its capacity for digital transformation.

32. **Subcomponent 2.3: Fostering regional trade facilitation.** Zambia is reforming its border operations to minimize interventions, aligned with ongoing initiatives under a Coordinated Border Management approach. This subcomponent will support Zambia's efforts to promote regional integration by providing TA aimed at increasing efficiencies at the border points and moving toward 'no-stop' borders along key digital smart corridors, in close cooperation with COMESA and neighboring countries. The project will support government efforts to help reform border operations and minimize interventions, facilitate cross-border data exchange, ensure mutual recognition of IDs, harmonize e-transactions, and enhance trade facilitation through automation reforms. The expected increase in regional trade is measured through the PDO indicator on regional data flows. COMESA is benefitting from a grant under IDEA Phase 1 and will support regional coordination with the Government, the Smart Africa Trust Alliance (SATA), and the Southern Africa Development Community (SADC) while leveraging ongoing trade initiatives supported by the World Bank projects such as Trade Facilitation Support Program (P163154; TFSP),³⁵ Zambia Agribusiness and Trade Project II (P179507; ZATP II),³⁶ and the

³³ See World Bank. 2017. *Cloud Computing Assessment and Action Plans*.

³⁴ SZI is officially called the Electronic Government Division formed by the Electronic Government Act, 2021.

³⁵ TFSP: <https://www.worldbank.org/en/programs/trade-facilitation-support-program>

³⁶ ZATP II: <https://projects.worldbank.org/pt/projects-operations/project-detail/P179507>



Transport Corridors for Economic Resilience (TRACER) project,³⁷ aiming to grow regional digital trade. This subcomponent, carried out in conjunction with COMESA and other IDEA countries, will include the following activities:

- (a) Conducting a feasibility study on the move toward no-stop borders. This includes automation reforms (single window, single payments, and so on), risk management, and one-stop (or no-stop) border solutions with neighboring countries to enhance trade facilitation in accordance with the World Trade Organizations Trade Facilitation Agreement.
- (b) Developing mechanisms to facilitate cross-border data exchange. The ZRA is already exchanging customs data with Botswana, Malawi, and Zimbabwe, and is currently working with DRC, Namibia, and Tanzania to develop customs system interconnectivity for electronic data exchange. ZRA is keen to enhance data sharing features and expand to other trading partners, notably South Africa, to gain advance notice of goods in transit.
- (c) Piloting cross-border mutual recognition of national IDs and ID verification mechanisms.
- (d) Harmonizing e-Transactions and cross-border regulations on data protection, privacy, and exchange with clear guidelines for cross-border data flows, processing, and storage of personal and sensitive data.

Component 3: Digital skills and capabilities (US\$5 million in IDA grant)

33. **This component focuses on building digital competencies at both foundational and advanced levels, specifically targeting underserved citizens** By providing basic digital literacy and developing employment-ready digital skills, this component aims to empower digitally enabled citizens, promoting the productive use of digital platforms and increasing the consumption of e-services. Additional support will be provided to develop advanced digital skills, thereby strengthening the supply of talented individuals equipped with market-aligned advanced digital skills to support the development of the Zambian ICT sector. Activities under this component are outlined in the following paragraphs.

34. **Subcomponent 3.1: Enhancing basic digital literacy.** Through SZI and ZICTA, the Government plans to deploy Digital Ambassadors within communities to provide foundational digital literacy training. SZI has already piloted basic digital skills training, leveraging content and platforms from the ITU and GSMA, in four Digital Transformation Centers (DTCs) located in schools, each equipped with internet connectivity provided via Starlink satellite technology. The expanded number of DTCs will serve as hubs, providing free internet access and offering training in digital literacy, entrepreneurship, and innovation to create job opportunities. In partnership with the Ministry of Youth, Sports, and Arts (MYSA) and Zambia Postal Services (ZAMPOST), SZI is working to expand this Program. This subcomponent will support the Government's efforts to scale up the basic digital literacy trainings, while ensuring online safety, targeting underserved populations at the base of the pyramid who are predominantly women, elderly, and PwDs. Additional digital literacy training will be provided by the Digital Ambassadors, particularly young women, targeting high-impact use cases such as using digital payment platforms and enhancing cybersecurity awareness within financial services, and for beneficiaries of social protection programs. This subcomponent will include the following activities:

- (a) Adapt a basic digital literacy curriculum and training manuals from best practice resources available while integrating local content translated into local languages and embedding cybersecurity awareness and considerations in the design of the training.
- (b) Provide basic targeted digital literacy training delivered through the DTCs in priority sectors, particularly targeting beneficiaries of social protection programs.
- (c) Expand the number of DTCs, and provide ICT equipment and renewable energy power supplies for them, and cover ICT equipment and operational costs for the Digital Ambassadors.

³⁷ TRACER project: <https://www.worldbank.org/en/news/press-release/2024/02/20/new-world-bank-grant-to-improve-transport-and-trade-connectivity-between-afe-zambia-and-tanzania> .



- (d) Promote productive digital usage and economic empowerment for women and girls by integrating gender- and disability-sensitive approaches (such as using assistive technologies), utilizing women trainers as Digital Ambassadors, including for sites connected under component 1. Support ZICTA and SZI in conducting digital skills and literacy education programs in targeted areas.
- (e) Develop a financial sustainability model for the DTCs to include private sector participation, and to offer trainees with vouchers for reduced cost training, to be redeemed at DTCs.
- (f) Facilitate knowledge exchange to learn from developed one-stop service centers such as the Kenya's *Huduma* centers and Rwanda's *Irembo* franchise model.

35. **Subcomponent 3.2: Nurturing employment-ready advanced digital skills.** This subcomponent seeks to develop a critical mass of skilled Zambian workers with advanced digital skills and capabilities, enabling them to harness the potential of emerging technologies. A DTC has been established at ZAMREN within the University of Zambia. This subcomponent will provide support to operationalize the ZAMREN DTC as a hub for advanced digital skills training. Training programs will be tailored to equip university faculty, and government officials, with advanced digital proficiency and students with readiness for the modern job market. To enhance the impact of the technical skills, trainings will be complemented with soft skills to improve collaboration, communication, problem solving, and so on. This subcomponent will include the following activities:

- (a) Conduct a comprehensive digital skills supply and demand gap assessment, with emphasis on advanced digital skills and future needs. This assessment will be based on interviews with industry stakeholders to determine the current availability of digital skills, the specific digital skills in demand by the private sector, and the skills currently being developed in universities and training centers. Additionally, the assessment will focus on the demand for remote workers for e-lancing and from global firms, identifying opportunities for Zambian workers to integrate into global digital value chains, and long-term tracking of those receiving training to track employability.
- (b) Provide training on advanced digital skills based on the gap assessment, at the DTCs, related to emerging technologies such as big data, AI, blockchain, and robotics. This training aims to strengthen the capacity of the Zambian economy to build local solutions and engage in global value chains, to leverage these technologies for economic development. Procedures for selection of trainees will be set out in the Project Implementation Manual (PIM). Training courses will be tailored to the beneficiaries, such as youth, women, elderly, teachers, civil servants, leaders etc.
- (c) Offer customized digital skills and online safety training for women, at the DTCs and via the Digital Ambassadors, along with advanced certified courses aligned with private sector needs. Incorporate rigorous assessment-based certification, mentorship programs featuring female tech role models, and community engagement initiatives to address social norms and foster inclusion.
- (d) Build digital skills and capabilities identified in the National AI Strategy.

Component 4: Program management and capacity building (US\$7 million in IDA grant)

36. **This component will support the critical building blocks for strong implementation, learning, and coordination of the Program.** It will finance the establishment of a Project Implementation Unit (PIU) within SZI and support project management and implementation of project-associated activities. This includes staffing and operating costs of the PIU, independent audits and quality assurance, and monitoring and evaluation (M&E) to support implementation and ensure compliance with Environmental and Social Standards (ESS). The overall project training plan will be included in this component.

C. Project Beneficiaries



37. DZAP will benefit all Zambians, including citizens, private sector businesses and government, targeting specifically public sector MDAs, women, and PwDs.

- **Individuals** including citizens and residents across Zambia, particularly women and those in rural communities, will benefit from expanded opportunities to participate in the digital economy through initiatives promoting digital access and inclusion. Some 7.4 million citizens ages 15 and above, at least half of whom are women and girls and particularly from rural communities, will benefit from new and enhanced internet connectivity, including from public access points. A further goal is to provide at least 2.5 million individuals with new digital IDs, enabling them to conveniently authenticate and verify their identities when accessing private sector and government services on ZamPortal and to make efficient digital payments through ZamPay. Additionally, at least 1,000 Zambians will receive advanced and specialized digital skills training.
- **Private sector businesses** will benefit from several key advances: (i) an increasingly digital-savvy and connected consumer base; (ii) investment in new DPI, such as improved connectivity, ID verification and authentication services, improved government-to-business (G2B) services, and improved access to public databases as a result of a data classification exercise; (iii) enhanced legal, regulatory, and institutional frameworks, including at the regional level; (iv) a stronger pool of digital talent; and (v) support for no-stop border posts. Direct beneficiaries will include mobile network operators, ISPs, wholesale fiber network operators, local IT equipment and service providers, private digital skills providers, and application services providers. They will directly benefit through competitively-awarded contracts for infrastructure deployment, internet prepurchase, local procurement of ICT services, and vouchers for discounted digital skills training. Digitally enabled businesses will benefit from increased access to markets through digital integration supported under the Program. The addition of at least 2,000 km of fiber optic connectivity in both middle mile and last mile networks will create new business opportunities, while efforts to improve the trading environment for cross-border digital services will foster enterprise and innovation.
- **Public sector institutions** will benefit from cost savings through the digitization and automation of core citizen-facing services, leading to improved service delivery. MDAs, particularly line ministries for ICT and trade, telecom and DFS regulatory bodies, and digital government agencies will benefit from this. The DPC and the Cybersecurity Authority will directly benefit from targeted financial support and TA. Unconnected government institutions, hospitals, health centers, local government authorities and public schools, connected under ZAMREN, particularly in rural areas, will directly benefit from last mile connectivity activities. These efforts will aim to connect 500 new locations, linking them to the GWAN and providing broadband access. Additionally, several MDAs will benefit from capacity building, with at least 500 public sector officials receiving specialized digital skills training.

38. The project focuses strongly on promoting digital inclusion and bridging the digital divide by empowering women and girls, PwDs, the elderly, and vulnerable communities in rural and underserved areas. Zambia's lowest income-earning households will benefit from initiatives aimed at providing connectivity access. Women, a particular focus of the project, are disproportionately affected by climate change, particularly in Zambia, where they account for 62 percent of agricultural labor,³⁸ with increasing extreme weather events affecting crops. ICT equipment and software applications financed by the project will include built-in accessibility features for PwDs, where appropriate. Public WiFi hotspots will be designed to ensure accessibility for women and PwDs, in gender-appropriate locations, and equipped with necessary assistive tools, such as screen readers and magnifiers. The project will also collect data disaggregated by gender and location on connectivity, digital skills, and access to services to monitor progress toward achieving digital inclusion targets.

³⁸ World Bank. 2025. World Development Indicators. <https://data.worldbank.org/indicator/SL.AGR.EMPL.FE.ZS?locations=ZM>.



D. Role of Partners

39. **DZAP will help bridge some of the private sector investment gap** aligned with MPA activities that have been designed to crowd-in private capital:

- **Private capital mobilization (PCM).** DZAP is expected to catalyze direct private sector investment in Zambia. For example, investment in additional backbone and last mile connectivity will directly leverage private capital co-investment, potentially with support also from IFC and MIGA. This will be achieved by using advanced procurement modalities, such as multi-round reverse auctions,³⁹ and bulk prepurchase of internet capacity, conducted over electronic auction platforms (see Box 1). Approximately US\$100 million in World Bank grant financing is expected to mobilize US\$20 million in UCF. A conservative approach was applied to the PCM calculations, with a ratio of public to private of 2:1 used for backbone and 3:1 for last-mile investments. While public-private sector financing ratios in comparative schemes in countries such as Tanzania (1:2) and the Comoros (1:1.5) are generally higher, conservative estimates have been used due to: (a) the low average revenue per user; (b) the impact of unstable foreign exchange rates and national shortages of foreign currency on investment costs in markets like Zambia; (c) lack of fully competitive markets and risk of licensing restrictions; and (d) low private sector appetite to invest capital up front within the time frame of project duration.
- **Private capital enabling (PCE).** DZAP is expected to facilitate private capital investments, of a similar order of magnitude to PCM, including through guarantees from MIGA; financing from IFC and other private investment institutions; and investments by telecom, ICT, and DFS companies toward the project's long-term outcomes. PCE is anticipated to be generated under DZAP through further market liberalization, growth in digital trade, and the creation of new investment opportunities, including for DPI. The resulting faster and cheaper networks will support innovation and additional investment from small- and medium-sized enterprises that will use these networks to develop new business ideas.

40. **The World Bank is well positioned to support continuous development of the digital sector, leveraging its technical expertise and history of advising the client and investing in key project areas.** For instance, the Digital Vice Presidency in the World Bank hosts dedicated and specialized teams in broadband, cybersecurity, ID4D, government-to-people digital payments, and Green Digital, which provide the client with opportunities to tap into expert knowledge and cutting-edge research. Project activities will leverage, enhance, and complement the existing efforts of other partners involved in supporting digital development in Zambia, aiming to capitalize on synergies and lessons learned. The World Bank has played a leading role in coordinating with donors and partners. For instance:

- The World Bank collaborates closely with UNICEF and ITU on initiatives such as GIGA, which seeks to map and enhance connectivity for schools;
- The Bank also works closely with the EU for the AfricaConnect program, now in its fourth phase, which aggregates demand for bandwidth and capacity building for RENS.
- The project's efforts to accelerate connectivity, DPI, and digital skills complement existing interventions by organizations such as the recent United States Agency for International Development (USAID) sector assessment⁴⁰ that has informed the technical design of project activities.

E. Lessons Learned and Progress on Learning Agenda

³⁹ World Bank, and CEPA. 2024. *Leveraging Private Sector Investment for Digital Communications Infrastructure in Eastern and Southern Africa*.

⁴⁰ USAID. 2022. *Digital Economy Country Assessment: Zambia*. https://www.usaid.gov/sites/default/files/2022-12/USAID_Zambia_DECA.pdf.



41. **As IDEA implementation is still in its infancy stages, while DZAP is being prepared, the learning agenda is still evolving.** Two pre-feasibility studies, supported by World Bank Trust Funds, were conducted to inform the project's design: a digital market integration regulatory study (2022) and an assessment of missing broadband links and IXPs in Southern Africa (2023). For Zambia specifically, a sector assessment published in 2020 informed project design,⁴¹ while an earlier study analyzed gender gaps.⁴² Through the MoHAIS and SZI, Zambia has benefited from TA from the World Bank's ID4D Trust Fund to accelerate an inclusive digital ID ecosystem by expanding ID enrollment and exploring sustainable ID credential options, such as digitally signed QR codes, plastic cards, or polycarbonate cards. Rolling out smart cards as the official ID credential is not envisaged as it would be costly and require significant infrastructure investments such as establishing smart card printing and production facilities, when less costly alternatives are available that offer similar efficiency.

42. **COMESA, as the regional coordinating body for IDEA, will play a central role in monitoring, evaluation, and developing the learning agenda.** The COMESA Secretariat, an experienced implementing agency for World Bank programs, is hosted in Zambia. It received a US\$10 million grant under IDEA Phase I and will leverage existing implementation structures and arrangements under current World Bank programs, such as ASCENT (P180547) and the Regional Infrastructure Investment Facility (RIFF, P171967). As far as possible, a coordinated approach will be pursued across these projects.

43. **The connectivity elements of DZAP will build on lessons learned from previous government connectivity efforts and a comparative analysis of last mile connectivity deployments elsewhere, to be commissioned under the PPA.** Additional insights will be drawn from implementation success of similar regional projects such as the Malawi Digital Foundations Project (P160533) and Digital Tanzania Project (P160766), which have made effective use of reverse auctions and bulk prepurchase of internet capacity based on long-term indefeasible rights of use (IRUs). Furthermore, the World Bank has extensive experience in working with countries to reform their telecom sectors and develop regulatory capacity in emerging markets, with recent and ongoing projects in countries such as Ghana, Kenya, Niger, and Rwanda.

44. **The project combines interventions that promote a whole-of-government approach to digitalization with investments that mitigate the rising risks associated with increased digital adoption.** Investments aimed at upgrading ID services and developing specialized digital skills for the industry draw on early lessons learned from countries such as Ethiopia, which is also progressing toward implementing a digital authentication layer based on an open-standards approach.

III. PROJECT IMPLEMENTATION

A. Institutional and Implementation Arrangements

45. As this is a new project in a country that has not benefited from any lending in the digital field for many years, a new PIU will need to be established. **Following discussions with the Government, a dedicated PIU for DZAP is being established and operationalized at SZI, using funds under the PPA.** Established in 2015, SZI is well suited to play this central role, given its mandate and track record of spearheading cross-cutting government digitalization initiatives. The functions of the PIU will cover project coordination, procurement, financial management (FM), environmental and social standards, M&E, communications, and gender. The project beneficiaries will include MoTS and SZI as implementing partners for Subcomponent 1.1; SZI, ZICTA, DPC, the CIRT and ZAMREN for Subcomponents 1.2 and 1.3; MoHAIS and MoTS as implementing partners for Subcomponent 2.1; SZI, MoTS and ZICTA for Subcomponent 2.2; the Ministry of Commerce,

⁴¹ World Bank. 2020. *Accelerating Digital Transformation in Zambia: Digital Economy Diagnostic Report*. <https://openknowledge.worldbank.org/handle/10986/33806>.

⁴² World Bank, and Research ICT Africa. 2017. *Gender and ICTs in Zambia*.



Trade and Industry (MCTI), MoTS, MoHAIS, and ZRA as implementing partners for Subcomponent 2.3; and ZAMREN, SZI, ZICTA, ZAMREN and MoTS for Subcomponents 3.1 and 3.2. Further details are provided in the PIM.

46. **The SZI PIU will lead implementation of the project's subcomponents and oversee core project-related fiduciary functions, including FM, procurement, project-related M&E, and environmental and social risk management.** It will also consolidate overall project reporting, working closely with all stakeholders. The PIU is being staffed using funds from the PPA with necessary experts, including a dedicated Project Coordinator and FM, procurement, environmental and social standards, and M&E specialists. Additionally, technical specialists in various project areas will liaise with core technical partners daily. Recruitment for most of the envisioned technical specialists and additional PIU staff is ongoing, with support from the PPA. The project will ensure that the hired technical experts transfer their knowledge and provide hands-on training to SZI staff to ensure sustainability after project completion. A few essential feasibility studies have also been commissioned under the PPA.

47. **SZI will closely collaborate with several MDAs as key stakeholders, technical partners, and main project beneficiaries for planned investments, and these MDAs have been actively involved in shaping the project design.** The project beneficiaries will include the MoTS, MoHAIS, SZI, DPC, CIRT, ZAMREN, MCTI, ZICTA, Ministry of Education and, later, Ministry of Health for connecting hospitals and health centers. These MDAs are expected to contribute technical inputs and evaluation of bids while working closely with the technical specialists embedded in the PIU. They will support SZI on technical aspects of procurement-related processes, support quality assurance and validation of key deliverables, and provide input needed for effective M&E framework management and reporting. Detailed roles and responsibilities and terms of engagement will be captured in the PIM, to be finalized as a condition of project effectiveness. Memoranda of Understanding (MoUs) will be established between the PIU and partner MDAs at the start of the project implementation, as needed.

48. **A dedicated Project Steering Committee (PSC), to be chaired by the Secretary to the Cabinet, will be established to provide strategic oversight and governance for the project,** with the National Coordinator of SZI serving as its secretary. The PSC's mandate will include: (a) approval of Annual Work Plan and Budget; (b) quarterly review of project progress; and (c) provision of strategic guidance and recommendations related to project implementation. The PSC will include representatives from key government institutions and project beneficiaries.

49. **The PSC will be supported by one or more targeted technical committees (TCs), as needed, to address any key *ad hoc* technical issues that arise and provide operational support.** These dedicated component-specific TCs will convene relevant government MDAs deemed essential for successful implementation and stakeholder management and support key technical-level decisions requiring broader agreement. TCs may include representatives from the private sector, academia, and civil society. TC meetings will be chaired and called by relevant SZI PIU technical specialists. TCs can escalate issues to the PSC if they require strategic discussion. The terms of reference for TCs, including expected membership, is detailed further in the PIM.

B. Results Monitoring, Evaluation, and Verification Arrangements

50. **DZAP's progress and implementation will be monitored not only at the level of the national projects but also at the regional level for the parent MPA (IDEA), where COMESA will play a coordinating role.** The IDEA Program has established PrDO-level indicators that will be applied and monitored across the Program, complemented by a Results Framework specific to each operation. Monitoring will be facilitated through the adoption of digital platforms, which will allow aggregation of results and individualized dashboards designed for stakeholders (World Bank, COMESA, government agencies, utilities, and the private sector) to improve continuous monitoring. For the PDO indicators, three key World Bank CRIs have been adopted, on internet access, digitally enabled services, and climate resilience. Among the PDO and intermediate indicators, 11 are gender tagged, 7 track PCM/PCE and enhancement, 4 measure climate co-benefits, while 4 monitor citizen engagement.



51. **The SZI PIU will be responsible for monitoring progress and achievement toward the PDO and intermediate indicators of the project's Results Framework detailed.** This requires staffing the PIU with an M&E expert to coordinate M&E centrally for all project activities across components and establishing an adequate M&E system based on the M&E plan, in coordination with COMESA. The status of project implementation will be documented in progress reports prepared on a semiannual basis and submitted to the World Bank, typically ahead of a mission, where reporting will be centralized, as far as possible. These reports will include updates on results, disbursements, FM, M&E, procurement, and ESS compliance.

52. **Built-in systems for tracking results and satisfaction surveys will be used to support citizen engagement and gather beneficiary feedback.** Related tools will be embedded directly into project delivery to ensure real-time feedback, using digital tools and systems to register beneficiaries and report their feedback. Additionally, beneficiary focus groups will be utilized to inform design and track progress over time.

C. Disbursement Arrangements

53. **The disbursement and funds flow will follow SZI procedures** as the existing systems that are in place are deemed adequate. However, there could be potential delays in the movement of funds from control 99, the Treasury Account, to the project operational accounts, thereby delaying project implementation. The project will open a US dollar Designated Account (DA), a kwacha operational account at the Bank of Zambia (BoZ), a US Dollar operational account and a 'mirror operational' kwacha bank account at a commercial bank acceptable to the World Bank. The project will use the statement of expenditure method of disbursement, whereby the DAs will have a fixed allocation of funds. Requests for replenishments and documentation of expenditures will be made at least once a month. SZI will request for advances using a withdrawal application in Client Connection and submit to the World Bank. The World Bank will disburse funds to the US\$ DA held at BoZ. To make payments in kwacha, SZI will move money from the US\$ DA to the operational kwacha account at BoZ and release payments for eligible expenditures through a mirror operational account at the commercial bank. Payments for eligible expenditure in US\$ will be made from the US Dollar operational account at a commercial bank. Other disbursement methods such as reimbursement, direct payment, and special commitment will be available on this project.

IV. PROJECT APPRAISAL SUMMARY

A. Technical, Economic and Financial Analysis

54. **DZAP is expected to contribute to sustainable economic growth, through long-term cost savings, efficiency, and productivity gains, fueled by greater digital adoption by citizens, businesses, and government.**⁴³ Improved accessibility and affordability of broadband services supported under Component 1 activities are expected to stimulate gross domestic product (GDP) and job growth in the country. Previous research suggests that a 10 percent increase in mobile broadband penetration in developing countries is associated with a 2.46 percent increase in GDP growth. The same research suggests that 10 percent increase in fixed broadband penetration in developing countries is associated with a 0.77 percent increase in GDP growth.⁴⁴ In addition, digital ID and public platforms activities under Component 2 and digital skills activities under Component 3 are expected to accelerate digital innovation and technology adoption among government public officials, citizens, and businesses. This will contribute to a holistic integration of the digital economy by the public sector as well as to increased competition, productivity, and consumer welfare among citizens and businesses, such as reduced travel requirements (Figure 2).

⁴³ World Bank. 2016. *World Development Report: Digital Dividends*.

⁴⁴ ITU. 2020. *How Broadband, Digitization and ICT Regulation Impact the Global Economy. Global Econometric Modelling*.



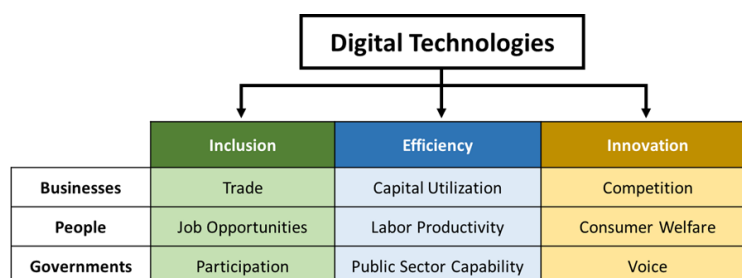
55. **The project-level economic and financial analysis undertaken follows a standard cost-benefit analysis approach and reveals a positive net present value (NPV) and internal rate of return (IRR).** The model relies on available secondary data and reasonable assumptions, based on experience and other projects in the digital portfolio, and additional evidence sourced from consultations, interviews conducted, and other similar interventions. The cost-benefit model was used to run a cashflow and financial analysis at the country level that features three different scenarios: Optimistic, Pessimistic, and Neutral. Cashflows are

expected to come from three main sources: economic impact in GDP growth due to broadband penetration increase, both fixed and mobile; economic impact due to digital services, digital ID and reduced KYC costs;⁴⁵ and economic impact from digital skills trainings and innovation. Discounted at 19.23 percent,⁴⁶ the Neutral scenario Net Present Value (NPV) reaches US\$14.57 million and yields a 33 percent Internal Rate of Return (IRR). For the Optimistic and Pessimistic scenarios, the NPV and IRR are US\$27.08 million and 45 percent and US\$0.62 million and 20 percent, respectively. It is worth noting that project impacts may go beyond the calculated estimated results, since there are some activities, especially those related to cybersecurity and regional trade, that could also produce economic and social gains. Those gains may come from the increase of online services and transactions that could occur from the improvement of safety and trust on the digital workspace at both the national and regional level. In addition, the reverse auction economic calculations for the 200 or so cellular sites expected to be deployed or upgraded under the project, and the 2,000 km of fiber expected to be deployed connecting 500 institutions, we estimate that US\$ 38 million in public funding will mobilize at least US\$ 20 million in private sector investment. This public-private contribution increases the financial viability of such interventions, rendering a higher IRR (13.2% without public investment and 23.8% with public investment) and a positive NPV (US\$ -18.5 million without public investment and US\$7.6 million with public investment). Furthermore, the GoZ can also expect an increase in VAT due to the growth of the subscriber base in rural areas, and job creation to operate and maintain the sites and fiber deployed.

56. **Paris alignment.** The project is fully aligned with the Paris Agreement on both mitigation and adaptation goals. **Assessment and reduction of mitigation risks.** The project will contribute to the country's low-carbon development goals by prioritizing renewable energy to power the digital connectivity infrastructure and digital ID infrastructure. It will also integrate energy efficiency measures in its physical infrastructure investments such as telecom equipment and digital ID biometric registration kits. The project's activities are considered universally aligned with mitigation goals since the integrated low-carbon measures will ensure that the carbon lock-in and transition risks associated with the project are low. The project will also help mitigate adaptation risks through investment in resilient digital communications infrastructure.

57. **Assessment and reduction of adaptation risks.** The main climate risks likely to have an adverse impact on project investments, as identified in climate and disaster risk screening (CDRS), are intense rainfall, floods, and high temperatures. They can damage telecommunication cables causing network failure in affected areas and inundate telecom equipment.

Figure 2. Digital Technology Benefits from Inclusion, Efficiency and Innovation



Source: World Bank. 2016. *World Development Report: Digital Dividends*.

⁴⁵ In India, eKYC resulted in costs being reduced by 90 percent, from US\$0.60 per customer to US\$0.07. Source: GSMA. 2016. "The Business Case for Payments Banks in India." <https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2016/12/GSMA-The-business-case-for-payments-banks-in-India-2016.pdf>

⁴⁶ The discount rate used for the calculations is 19.23 percent, which includes a spread reflecting the project risk and nuances.

Source: https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html



Droughts also contribute to unstable electricity supply. The project will integrate climate risk assessments in the design of digital connectivity infrastructure and accordingly embed structural climate adaptation measures for climate-resilient digital connectivity. It will also digitize paper government records for digital ID and births and deaths registration, as well as providing backup options (by supporting migration to cloud) to prevent data loss in the event of a climate-related emergency, such as flooding. The project is also designed to improve resilience by providing internet connectivity to underserved populations in climate-vulnerable hotspots and investing in digital ID, which will allow for quick identify verification for cash transfers to drought-impacted population. Overall, the abovementioned adaptation measures reduce the risks from climate hazards in this operation to an 'acceptable' level.

B. Fiduciary

(i) Financial Management

58. **FM assessment.** The World Bank carried out an FM assessment of the proposed SZI PIU in January 2025. The assessment was carried out in accordance with World Bank Guidance: FM Manual for World Bank IPF operations, September 2021, and World Bank Policy and Directive on IPF, February 2017. The purpose of the assessment was to evaluate the adequacy of FM arrangements in place at SZI to support project implementation. In carrying out the assessment, various pertinent documents were reviewed, a self-assessment questionnaire was administered, and interviews conducted with SZI staff to verify the adequacy of SZI's FM arrangements comprising of planning and budgeting, accounting, funds flow, internal controls (including internal audits), financial reporting, and external audits. The overall initial FM risk rating of the DZAP was assessed as *Substantial* with the risk after implementation of the mitigation measures assessed as *Moderate*.

(ii) Procurement

59. **Procurement activities, including approvals and payments for the project, will be coordinated and processed centrally by SZI.** Actual procurement processing and contract management will be implemented by the individual procuring entity, namely SZI. Both SZI and MoTS have been assessed to determine their experience and capacity to carry out procurement processes under World Bank-financed projects. As necessary, staff assigned to procurement implementation roles will be trained by the World Bank. A procurement capacity assessment of SZI, as the PIU that will be responsible for carrying out procurement of the project, has been performed using the World Bank's online Procurement Risk Assessment and Management System (PRAMS). The project procurement risk is rated as *Substantial*. This is because SZI has no prior experience in implementing World Bank-funded projects. SZI will be required to set up institutional systems and processes, ensuring that procurement complies with the World Bank's *Procurement Regulations for Investment Project Financing (IPF) for Borrowers for Goods Works, Non-Consulting and Consulting Services 5th Edition* September 2023 and the World Bank's *Guidelines for Preventing and Combating Fraud and Corruption* revised as of July 2026. When procurement with cost estimates below prior review thresholds is undertaken using National Bidding Procedures based on the use of the Zambia Public Procurement Act of No.8 of 2020 as amended by Public Procurement Amendment Act No 17 of 2023 and the accompanying Procurement Regulations of 2022, adjustments will be made to make the procurements acceptable to the Bank. Specifically, provisions of the Banks Procurement Regulations paragraphs 5.3 to 5.9 will be complied with in addition to eligibility, conflict of interest and application of the Bank's Guidelines for Preventing and Combating Fraud and Corruption.

C. Environmental, Social and Legal Operational Policies



60. **The project will have a nationwide footprint, with efforts targeted at expanding access to affordable and inclusive internet and digitally enabled services.** Through the project, national and regional networks will be expanded, including last mile connectivity for at least 500 selected national sites which will include schools, HEIs and, later, health facilities, among others. The project has a pro-green approach focused on strengthening regulation for the management of e-waste and applying a dig-once policy to reduce negative impacts on communities and the environment. The project has adopted an inclusive approach that is intended to ensure that project benefits accrue to residents across the country, including rural areas.

61. **An assessment of the project reveals potential environmental and social risks and impacts and occupational health and safety (OHS) risks that may result from proposed activities.** These could include: (a) excavation of trenches for installation of fiber optic cables; (b) minimal vegetation loss; (c) generation of e-Waste; (d) exposure to electrical hazards, including working under high-voltage supply lines; (e) traffic-related incidents; (f) transmission of communicable diseases; (g) potential exclusion of vulnerable groups from accessing project benefits; (h) potential incidences of sexual exploitation and abuse or sexual harassment (SEA/SH) due to contractor-community interactions; (i) potential loss of data privacy due to exposure of personal information to unauthorized individuals; and (j) theft of fuel or oil and solar equipment powering ICT infrastructure if not properly secured. Measures to mitigate these risks have been incorporated in the Environmental and Social Management Framework (ESMF), which was disclosed by SZI on its website on November 29, 2024.⁴⁷ Site-specific Environmental and Social Management Plans (ESMPs), including measures to manage SEA/SH risks, Labor Management Procedures (LMP), and the provisions of the OHS matrix incorporated into the ESMF, aligned with the World Bank Environmental, Health Safety Guidelines and Good International Industry Practices, will be developed. The project has prepared and disclosed a Stakeholder Engagement Plan (SEP)⁴⁸ and an Environmental and Social Commitment Plan (ESCP)⁴⁹ with actions and timelines for achieving compliance, approved on February 7, 2025. TA activities involving capacity building are likely to generate downstream environmental and social risks and impacts. Mitigation measures will include reviewing TORs and Annual Work Plans to ensure activities are conducted in line with the Environmental and Social Framework (ESF) requirements and using alternative technologies, such as satellite, where trenching is not required.

Legal Operational Policies	Triggered?
Projects on International Waterways OP 7.50	No
Projects in Disputed Area OP 7.60	No

V. GRIEVANCE REDRESS SERVICES

62. **Grievance redress.** Communities and individuals who believe that they are adversely affected by a project supported by the World Bank may submit complaints to existing project-level grievance mechanisms or the Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and individuals may submit their complaint to the Bank's independent Accountability Mechanism (AM). The AM houses the Inspection Panel, which determines whether harm occurred, or could occur, as a result of Bank non-compliance with its policies and procedures, and the Dispute Resolution

⁴⁷ https://www.szi.gov.zm/wp-content/uploads/2024/11/DZAP_Environmental-and-Social-Management-Framework_Draft_28.11.2024.pdf

⁴⁸ https://www.szi.gov.zm/wp-content/uploads/2024/11/DZAP-Stakeholder-Engagement-Plan_Draft.pdf

⁴⁹ https://www.szi.gov.zm/wp-content/uploads/2024/11/DZAP-Environmental-and-Social-Commitment-Plan_28.11.2024.pdf



Service, which provides communities and borrowers with the opportunity to address complaints through dispute resolution. Complaints may be submitted to the AM at any time after concerns have been brought directly to the attention of Bank Management and after Management has been given an opportunity to respond. For information on how to submit complaints to the Bank's Grievance Redress Service (GRS), visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the Bank's Accountability Mechanism, visit <https://accountability.worldbank.org>.

VI. KEY RISKS

63. **The overall risk for DZAP is *Substantial*.** Risks rated *Substantial* or higher are elaborated below:

- **Macroeconomic risk is *Substantial*,** as the country is classified as being in debt distress. The country's debt default coincided with the COVID-19 pandemic and it is not immune from similar shocks in the future, particularly when the current grace period on principal repayments and negligible interest rates expires after 2025. The macroeconomic situation may deter potential investors. To counter these risks, the project does not rely at all on counterpart financing from the Government and the proposed financing is entirely grant based.
- **Sector Strategies and Policies risk is *Substantial*.** The main risks here are related to digital ID implementation -- including those related to exclusion, data protection and privacy, and to unsuitable or unsustainable technology choices. The Government's commitment to a "home-grown enrolment" system does not necessarily guarantee enhanced privacy or security. Nevertheless, the current policy direction towards open standards is encouraging and World Bank TA, provided through the 2024 ID4D diagnostic, is helping to avoid potential pitfalls. Further support on preparing data protection guidelines for digital ID is being provided through the PPA. A secondary risk may arise if the private sector does not co-invest in infrastructure and requires 100 percent subsidy from project funds. This would result in a lower level of infrastructure investment but would not invalidate the principles of private sector ownership and operation of network on which the project is founded. The risks can be mitigated through a competitive and transparent bidding process for subsidies for communications infrastructure provided through project funds.
- **Institutional capacity for implementation and sustainability is *Substantial*,** largely because this is a new project with an untested PIU. Nevertheless, SZI has been in existence for almost a decade now and has gained considerable experience in working on development projects with the donor community, notably with USAID, Japan International Cooperation Agency (JICA),⁵⁰ and United Nations Development Programme (UNDP).⁵¹ The World Bank Country Office completed FM and procurement assessments with relevant focal points in SZI and MoTS, in June 2024, and appropriate risk ratings have been adopted. The experience gained from implementation of the PPA will help the PIU get established. Other mitigation steps include involving COMESA in developing TORs and project monitoring. Specifically, the PPA will be used to further develop data protection regulations relevant to rollout digital ID.
- **Fiduciary risks are *Substantial*** for much the same reason, namely because this is a new PIU with no previous experience of implementing World Bank programs. The risk ratings for both FM and procurement are initially set as '*Substantial*' but are expected to be reduced to '*Moderate*' following the implementation of the risk mitigation measures set out in the PIM. Additional training will be provided in the use of the Bank's Systematic Tracking of Exchanges in Procurement (STEP) and Client Connection.
- **Other risks, which include Cybersecurity and Climate, are also *Substantial*.** Two new Acts approved in April 2025, dealing with Cybersecurity and Cybercrimes, have raised some concerns among donor partners and

⁵⁰ JICA project for enhancement of science and mathematics education at secondary schools.

⁵¹ UNDP National Innovations Initiative / Accelerator Labs in Zambia.



civil society over the extension of its scope, and possible challenges to freedom of expression and data protection. To mitigate these risks, Bank TA will focus on cybersecurity rather than cybercrimes. On the climate side, the recent severe drought is still having a negative effect on stability of electricity supply, and this is impacting the provision of digital services. To mitigate this risk, the project will focus investment on increasing the supply of renewable solar energy.



VII. RESULTS FRAMEWORK AND MONITORING

PDO Indicators by PDO Outcomes

Baseline	Closing Period
Increase access to, and inclusive use of, the internet	
People using broadband internet (new use) (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	3,400,000
➤ People using broadband internet (new use) - Female (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	1,700,000
➤ People using broadband internet (new use) - Youth (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	571,000
➤ People using broadband internet with a fixed-line connection (new use) (Number)	
Nov/2022	Oct/2030
0	700,000
People using broadband internet (inferred use) (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	2,000,000
➤ People using broadband internet (inferred use) - Female (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	1,000,000
➤ People using broadband internet (inferred use) - Youth (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	336,000
People using broadband internet (enhanced use) (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	2,000,000



➤ People using broadband internet (enhanced use) - Female (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	1,000,000
➤ People using broadband internet (enhanced use) - Youth (Number of people) ^{CRI}	
Nov/2022	Oct/2030
0	336,000
Volume of international data traffic: used international bandwidth in kbit/s per capita (Number)	
Nov/2023	Oct/2030
29.80	90.00
Increase access to, and inclusive use of, digitally enabled services	
People using digitally enabled services (new services) (Number of people) ^{CRI}	
Nov/2021	Oct/2030
0	2,000,000
➤ People using digitally enabled services (new services) - Female (Number of people) ^{CRI}	
Nov/2021	Oct/2030
0	1,000,000
➤ People using digitally enabled services (new services) - Youth (Number of people) ^{CRI}	
Nov/2021	Oct/2030
0	336,000
People using digitally enabled services (enhanced services) (Number of people) ^{CRI}	
Nov/2021	Oct/2030
0	5,000,000
➤ People using digitally enabled services (enhanced services) - Female (Number of people) ^{CRI}	
Nov/2021	Oct/2030
0	2,500,000
➤ People using digitally enabled services (enhanced services) - Youth (Number of people) ^{CRI}	
Nov/2021	Oct/2030
0	840,000
People with digitally-verifiable identification (Number)	
Nov/2022	Oct/2030
1,500,000	4,000,000
➤ Females with digitally verifiable identification (Number)	
	Oct/2030



2,000,000

Intermediate Indicators by Components

Baseline	Closing Period
1. Affordable broadband	
People benefiting from climate resilient infrastructure (Number of people) ^{CRI}	
Dec/2022	Oct/2030
0	1,000,000
➤ People benefiting from climate resilient infrastructure - Female (Number of people) ^{CRI}	
Dec/2022	Oct/2030
0	500,000
➤ People benefiting from climate resilient infrastructure - Youth (Number of people) ^{CRI}	
Dec/2022	Oct/2030
0	162,800
Retail price of a standard package of mobile data services per month, as a share of GNI per capita (Percentage)	
Nov/2024	Oct/2030
6.40	2.00
Kilometers of fiber optic network added (Kilometers)	
Nov/2024	Oct/2030
0	2,000
➤ Of, which, climate resilient and energy efficient fiber (Kilometers)	
Nov/2022	Oct/2030
0	1,600
Public institutions provided with new or improved access to broadband (Number)	
Nov/2023	Oct/2030
0	500
Average broadband speed for a fixed-line internet user, in Mbit/s (Number)	
Oct/2024	Oct/2030
17.18	50.00
2. Interoperable and secure platforms	
Number of entities on-boarded onto the government data exchange platform (ZamConnect) (Number)	



Dec/2023	Oct/2030
0	15
Government data governance frameworks, including data classification and cloud policies, enacted and mandated for use (Number)	
Dec/2023	Oct/2030
0	1
NUmber of monthly transactions handled by Government data exchange platform (Zamportal) (Number)	
Dec/2023	Oct/2030
0	10,000
People who have made or received a digital financial payment (Number)	
Dec/2021	Oct/2030
8,956,200	10,000,000
➤ Females who have made or received a digital financial payment (Number)	
Dec/2021	Oct/2030
4,238,400	5,000,000
3. Digital Skills and Capabilities	
Graduates of advanced digital skills training programs (Number)	
Nov/2023	Oct/2030
0	1,000
➤ Female graduates of advanced digital skills programs (Number)	
Nov/2023	Oct/2030
0	500
Number of Digital Transformation Centers operating (Number)	
Oct/2024	Oct/2030
4	8
Number of Digital Ambassadors appointed (Number)	
Oct/2024	Oct/2030
0	100
➤ Number of female Digital Ambassadors appointed (Number)	
Oct/2024	Oct/2030
0	50
4. Program management and capacity building	
Private capital mobilized (Amount(USD))	
Nov/2023	Oct/2030



0	20,000,000
Citizen engagement indicator: Grievances registered that receive an adequate response within 30 days (Percentage)	
Nov/2023	Oct/2030
0	80



Monitoring & Evaluation Plan: PDO Indicators by PDO Outcomes

Increase access to, and inclusive use of, the internet	
People using broadband internet (Number of people) ^{CRI}	
Description	The number of people and businesses using new or enhanced internet broadband facilitated through project support. Broadband internet is defined as 3G or higher, with a minimum download speed of 256 kbps. Use is measured as the number of subscribers to broadband internet and is disaggregated by sub-indicators (New Use, Inferred Use, and Enhanced Use), gender, youth, and by fixed/mobile users.
Sub-indicator: New users	The number of people who have become new subscribers to broadband internet, when they had not been in the past, as a result of last mile investments and interventions, such as connecting households, villages, schools, health centers, businesses, or other facilities.
Sub-indicator: Inferred use	The number of people who have become new subscribers to broadband internet as a result of investments and interventions at the first or middle mile/upstream segments of the value chain or through regulatory changes that improve the enabling environment, with the impact occurring further down the results chain.
Sub-indicator: Enhanced use	The number of people with existing subscriptions that have benefitted from improved broadband internet. Improved use refers to enhanced download and upload speeds as well as lower prices for data plan subscriptions.
Frequency	Annual
Data source	GSMA, ZICTA
Methodology for Data Collection	This indicator is mainly based on subscriber numbers, as reported by operators to the regulator/GSMA, but with multipliers applied, for instance, to take account of number of users in a typical school, HEI, government office, and so on, provided with broadband internet. These multipliers are obtained from survey data. As internet in schools is targeted, internet users from age 5+ are considered.
Responsibility for Data Collection	PIU
Increase access to, and inclusive use of, digitally enabled services	
People using digitally enabled services (Number of people) ^{CRI}	
Description	The number of people and businesses using new or enhanced digitally enabled services supported by project activities and provided by the public or private sector. Digital services are those that facilitate a targeted two-way digital interaction (data transmission or exchange between provider and end user) through external networks such as internet or SMS with users.
Sub-indicator: New services	The number of users of services that were entirely manual in the past but are now digitalized (partially or fully).
Sub-indicator: Enhanced services	The number of users of services that might have been digitalized partially or fully in the past but have now improved or deepened the digitalization to make those services more efficient and affordable for the end user.
Frequency	Annual
Data source	Project data and Findex survey
Methodology for Data Collection	This indicator is based on supply-side data from ministries/operators providing digitally enabled services, backed up by demand-side data from usage surveys.
Responsibility for Data Collection	PIU, BoZ



ANNEX 1: Climate

Climate vulnerability profile: Zambia is frequently inundated with seasonal floods and flash floods and affected by extreme temperatures and droughts, with increased severity and frequency in recent decades. Inadequate infrastructure, paired with the fact that a large proportion of the population is rural and poor, makes Zambia highly vulnerable to climate change and natural hazards, especially floods.⁵² The mean temperature in Zambia is expected to increase by 1.69°C by mid-century. The number of consecutive dry days is expected to increase by 25–49 percent between the middle and end of the century.⁵³ These shifts indicate a trend toward drier conditions overall, with potential for increased intensity during rainfall events. Consequently, climate hazards will increasingly have devastating consequences, exacerbating Zambia’s development challenges, as described in the NAP. Zambia is currently experiencing the driest agricultural season in more than 50 years, resulting in significant crop losses, increased livestock deaths, and worsening poverty.⁵⁴ Based on a high-level risk assessment, Zambia’s existing and planned digital connectivity infrastructure is highly vulnerable to damage from intense rainfall, floods, and high temperatures. Flooding can damage critical infrastructure, including disrupting the country’s connectivity/broadband internet network for extended periods of time. While damage may be localized, the impact can be at the level of the entire network.

Climate adaptation and mitigation activities under the Zambia project

Component 1: Affordable Broadband Adaptation

US\$50 million
IDA

For both subcomponents 1.1 and 1.2, Network design will be informed by site-specific climate risks identified in the climate risk assessments (to be conducted by the operator and built into the tender). Adequate adaptation measures will be embedded to ensure the new connectivity infrastructure is climate resilient.⁵⁵

Both subcomponents 1.1 and 1.2 will provide financing for broadband coverage that will enable climate/drought-impacted communities to easily and rapidly access drought Electronic Cash Transfers (ECTs) and improve their resilience to droughts.

Areas to be covered under this component will prioritize climate hotspots⁵⁶ and specifically target areas with rural, traditionally marginalized populations most severely and frequently affected by high levels of droughts (such as communities in Luapula,⁵⁷ Muchinga, and elsewhere) where digitally enabled responses to climate events are limited both by a lack of climate-resilient digital infrastructure and poor digital connectivity,

⁵² The country ranks 132 out of 185 countries included for 2021 on the ND-GAIN Index.

⁵³ Data based on World Bank’s Climate Change Knowledge Portal.

⁵⁴ Over 9 million people in 84 out of the 117 districts in Zambia are affected by drought according to the Government crop assessment data (<https://www.unocha.org/publications/report/zambia/zambia-drought-response-appeal-may-2024-december-2024-may-2024>). The drought is expected to create a 430 MW power deficit and affect ground and surface water levels, affecting sectors beyond agriculture, as over 80 percent of Zambia’s electricity is generated from hydropower.

⁵⁵ Climate resilience measures for the broadband connectivity infrastructure will follow international best practice, including recommendations from ITU-T on adaptation (Recommendation ITU L.1502) to determine the choice of technology, such as between underground and aerial fiber optic cables, and deploy weather-resistant fiber optic, weather-proofing the ducts, poles, switches, sockets, and appliances in the network, embedding elevation in the communication towers and water-sensitive equipment to prevent damage from floods and heavy precipitation.

⁵⁶ Climate- vulnerable hot spots are those that are often subject to droughts and/or floods and have limited access to digital connectivity and digital services

⁵⁷ CGIAR. 2022. *New Hotspot Mapping Reveals Where Climate Change Hits Women the Hardest*. <https://www.cgiar.org/news-events/news/new-hotspot-mapping-reveals-where-climate-change-hits-women-the-hardest/#:~:text=We%20found%2C%20for%20example%2C%20that,area%20prone%20to%20seasonal%20flooding>.



hampering the Government's response to climate-related emergencies⁵⁸. This is especially intended to increase efficiencies in cash transfer programs through the use of mobile payments in the drought-impacted areas⁵⁹. The selection of communities to be covered under this component will be determined during implementation, based on an assessment of two key criteria: un- or under-served by affordable/quality broadband internet service, and high severity and frequency of drought impact⁶⁰. This will inform the design and financing of the connectivity under this component to address the coverage and financing gap and deliver broadband internet coverage in the identified areas.

"Enhanced social protection programs with improved targeting of beneficiaries", specifically for rural populations, is an adaptation priority identified in Zambia's National Adaptation Program (NAP) that this project will support by directly contributing to the Government's SCT program that aims to provide cash transfers to the 84 drought-affected districts. Mobile transfers are a more reliable way to deliver drought response electronic cash transfers (ECTs), but a key barrier identified is that they're not possible in very remote areas⁶¹ – which the connectivity financed under this project will directly address by specifically targeting those areas (i.e unconnected populations in drought-affected districts). This will enable the Government to quickly deliver these time-bound emergency transfers ahead of, during, or in the aftermath of a climate event and directly contribute toward implementing Government's plan to deploy the digital system countrywide.⁶².

Zambia's national digital-based SCT payment system is being utilized to support government's drought response and relief efforts. Additionally, the Government recently completed enumeration of additional drought-impacted households nationwide for temporary emergency assistance through the SCT program during the drought period⁶³. Consequently, network connectivity expansion under this component will allow existing beneficiaries (and cover more beneficiaries) in these areas to easily and rapidly access drought ECTs and improve their resilience to droughts.

Mitigation

All connectivity-related activities financed under Component 1 will leverage the use of energy efficiency technologies and measures to support energy conservation and GHG emissions. These activities will meet best international practices for energy efficiency and energy performance⁶⁴ and green ICT standards,⁶⁵ to the extent technically and financially feasible. All telecommunications network equipment to be procured under this component will have energy efficiency levels that will surpass the country's current energy efficiency levels and equipment procured for similar expansion projects. There are currently no mandatory minimum energy

⁵⁸ As of May 2024, about 84 out of Zambia's 117 districts were affected by a drought that was considered the worst in at least two decades. The drought began in January 2024 and affected the Central, Copperbelt, Eastern, Lusaka, Northwestern, Southern, and Western provinces. On February 29, 2024, Zambia declared a national emergency due to the drought, which was caused by climate change and El Niño. <https://reliefweb.int/disaster/dr-2024-000018-zmb>

⁵⁹ Expanded connectivity under this project would allow these beneficiaries as well as government offices and stakeholders responsible for social protection program in climate- vulnerable hot spots, to remain connected at the time of crisis and ensure digitally enabled delivery of basic services, for example, emergency cash transfers (ECTs) through mobile money

⁶⁰ The project will conduct assessment/study in the implementation stage to identify communities/areas with high severity & frequency of drought (climate hotspots) that are unserved or underserved by affordable and quality broadband internet services.

⁶¹ SCT involves small frequent payments, making it difficult for beneficiaries to travel long distances every time, and the SCT guidelines specify that pay points should be no further than 7 km away from communities. This means that digital payments for SCT will not be possible everywhere until network connectivity and private sector participation improve in the most remote areas of the 84 drought-impacted districts. World Bank. 2022. *Digitizing Cash Transfers to Remote Rural Populations: Challenges and Solutions from the Experience of Zambia*.

⁶² <https://documents1.worldbank.org/curated/en/099131507172430988/pdf/IDU103b1adfa1d5db148ad19dac17f1c4c2a1a3a.pdf>

⁶³ Second Additional Financing for the Scaling-up Shock Responsive Social Protection Project (P181711).

⁶⁴ Canfora, P., P. Gaudillat, I. Antonopoulos, and M. Dri. 2020. *Best Environmental Management Practice in the Telecommunications and ICT Services Sector*. EUR 30365 EN, Publications Office of the European Union, Luxembourg.

⁶⁵ For instance, ITU-T, *Green ICT Standards and Supplements*. <https://www.itu.int/net/ITU-T/lists/standards.aspx?Group=5&Domain=28>.



performance standards and energy labeling system in Zambia for network equipment and connectivity deployment. ⁶⁶	
1.1 Extending the backbone and regional networks	US\$20 million
Adaptation Demand aggregation for internet capacity at the regional level will contribute to reduced cost of internet in communities and centers located in climate-vulnerable hotspots (see description above under component 1 in the climate annex) Mitigation The project will deploy fiber optic technology for backbone and regional networks, which is the best available technology for energy-efficient infrastructure deployment in the country (see description under Component 1). Extending fiber backbone coverage for backbone and regional networks will help reduce the size of the microwave backhauling network (currently, microwave networks are used extensively in Zambia for backbone connectivity and last mile). This will result in a significant reduction in GHG emissions associated with the data traffic as optical fiber is two to three times more efficient than microwaves for backhauling, allowing power savings from 54 to 94 percent on the backhaul deployment in a 4G scenario.⁶⁷ The equipment procured will be energy efficient (see description under Component 1).	Expected US\$20 million fiber optic
1.2: Supplying last mile connectivity Adaptation As noted above, the network design and network configuration for digital connectivity will be informed by site-specific climate risks identified in the climate risk assessments. Under the competitive bidding process, a key prioritization criterion will be to connect climate-vulnerable hot spots sites that are in un- or under-served by affordable/quality broadband internet service to enable and extend electronic cash transfers to drought impacted beneficiaries of the SCT (See description under component 1 in the climate annex). Additionally, LEO satellite connectivity deployed under this subcomponent will be used for emergency responses and will strengthen climate resilience and allow subscribers to receive emergency alerts.⁶⁸	US\$25 million Expected US\$10 million fiber; US\$10 million satellite; US\$5 million cellular

⁶⁶ <https://rise.esmap.org/country/zambia>

⁶⁷ <https://europacable.eu/wp-content/uploads/2022/07/Politecnico-di-Milano-for-Europacable-Energy-efficiency-of-fiber-on-mobile-networks-December-2021-2.pdf>.

⁶⁸ The results framework includes the corporate scorecard indicator on the increase in the number of people benefitting from climate-resilient infrastructure, broken down by gender and youth.



Mitigation Expectations are that 20 percent (US\$5 million) of investment will comprise new and improved cellular (an upgrade from 2G cell sites to more energy-efficient 4G/5G cellular technologies⁶⁹), 40 percent (US\$10 million) new satellite and/or mix of LEO satellite⁷⁰ and new 4G/5G cellular cell sites, with the remaining 40 percent (US\$10 million) new and improved fiber,⁷¹ depending on the outcomes of the competitive tendering process. Similar to Subcomponent 1.1, the equipment procured will be low carbon and energy efficient, which will surpass country's current energy efficiency levels and equipment procured for similar expansion projects. This includes adopting green (low-carbon) routing, using renewable (solar) energy sources to power base transceiver stations or other equipment to the extent possible in rural and remote areas with limited or no access to grid electricity, and switching off routers and selective interfaces during low demand.	
1.3: Strengthening the enabling environment	US\$5 million
Adaptation and mitigation To minimize GHG emissions from e-waste disposal, the project will finance climate-informed management of e-waste as stipulated in the national e-government plan. The development of e-waste regulations, guidelines, and disposal facilities will primarily integrate and focus on reuse, repair, and recycle.	US\$0.5 million
Component 2. Interoperable and secure data platforms	US\$38 million
2.1: Developing digital ID services	US\$23 million
Adaptation The identity verification services supported under this project will streamline the delivery of social protection packages to beneficiaries, including those affected by climate shocks (see SCT under Subcomponents 1.1 and 1.2). The activity will improve the Government's ability to target beneficiaries of the SCT program⁷² and enable digital ID verification for cash transfer that requires proof of vital events including in cases following drought (also see description under Component 1 for climate-vulnerable hotspots that will benefit from quick and timely electronic cash transfers as a result of digital access provided under this project). Mitigation Procurement of all ICT equipment worth around US\$20 million for the upgraded digital ID system, including solar-powered biometric registration kits (costing around US\$7 million), other ICT equipment, including laptops, and printers, will leverage high energy-efficient technologies and measures to support energy conservation and	US\$23 million = US\$7 m (solar-powered biometric registration kits) + US\$13m (remaining ICT equipment for digital ID system) + US\$3m (operational activities such

⁶⁹ GSMA and World Bank report that energy efficiency improves severalfold from 2G/3G to 4G/5G, thus encouraging investments in and migration from 2G/3G to 4G/5G to improve efficiency and reduce both energy consumption and carbon emissions per unit of data. GSMA. 2023. *Going Green: Benchmarking the Energy Efficiency of Mobile Networks* (second edition) and World Bank. 2024. *Green Digital Transformation: How to Sustainably Close the Digital Divide and Harness Digital Tools for Climate Action*. *Climate Change and Development Series*.

⁷⁰ LEO satellite installations would, most likely, be fully solar powered as not only do the orbiting satellites use solar power but also the ground segment of small receivers can be powered by solar arrays, with battery support. Moreover, LEO satellite connectivity serves as a robust supplement or alternative to terrestrial fiber and mobile cell sites in remote areas and climate-vulnerable hotspots highly prone to climate hazards such as floods, storms and droughts, ensuring reliable and redundant communication and data transfer where traditional infrastructure is sparse and may be compromised, thereby enhancing overall resilience against climate-related disruptions to digital connectivity. Satellite systems are relatively immune to the physical damage that often affects ground-based networks during severe climatic hazards.

⁷¹ Any new 4G/5G cell sites funded under the project would be fully solar-powered with battery storage.

Without the World Bank's intervention, and in a non-project scenario, the use of energy-intensive microwave (instead of the modern energy-efficient fiber optic cables) for broadband connectivity would be far more extensive in Zambia.

⁷² Zambia's Government issued a Drought Response Appeal in May 2024 to help fill the estimated critical financing gap that covers the SCT program noted above. Following this, the UN has worked with the Government and cooperating partners to prepare a UN drought response plan, issuing over US\$228.3 million for 67 projects in food security; agriculture; health and nutrition; education; water, sanitation, and hygiene; and social protection.



reduce GHG emissions. This will be achieved using standards noted above, including Energy Star-certified ICT equipment, which would surpass the energy efficiency levels of the equipment existing in the country.⁷³ Physical and mobile ID enrollment sites will be powered with renewable energy solutions, to the degree possible, reducing reliance on diesel-powered generators.	as transport, and enrollment)
2.2: Enhancing digital government	US\$13 million
Adaptation This subcomponent will support digitization of paper-based records for Digital ID and births and deaths registration to reduce the risks of record loss in the event of flooding. Targeted sites exposed to higher climate risks will be prioritized. Developing a data classification and governance framework with guidelines to migrate legacy records to digital format, including migration to cloud-based services. Paper-based records are stored locally across the country, with no aggregated backups in a centralized repository. This makes them vulnerable to damage or loss during climate disasters such as floods.⁷⁴ This risk could be reduced by digitalizing critical paper records such as birth and death registration and backing them up electronically. The project will support migration to cloud-based services at the national and regional levels, especially of sensitive government data and records, as well as backup options to prevent data loss in the event of climate-related emergency, such as flooding. This is aligned with Zambia's National Electronic Government Plan (2023–2026) that explicitly includes the activity “establish disaster recovery and backup systems at the existing Government data center” and aims to implement strategies and projects for reducing paper documents in public bodies, which will lead to improved climate change resilience in digital government, enabling improved government business continuity and climate disaster response. Mitigation Similar to Subcomponent 2.1, procurement of all ICT equipment for digital government (estimated at US\$10 million) will be certified by Energy Star or equivalent standards, which would surpass the energy efficiency levels of equipment existing in the country (see description under Component 1).	Purchase of energy-efficient ICT equipment and services, around US\$10 million
2.3: Fostering regional trade facilitation	US\$2 million
Mitigation Greater use of digital data transfer between countries in the region will reduce the requirement for physical transfer of goods, for instance digital streaming of music replacing sales of vinyl and compact discs.	
Component 3: Digital skills and capabilities	US\$5 million
3.1: Enhancing basic digital literacy Training programs in basic digital literacy, to be delivered through the DTCs and facilitated by the Digital Ambassadors, will incorporate green perspectives by promoting environmentally sustainable practices. These programs will emphasize topics such as energy-efficient use of digital devices, responsible e-Waste management to reduce GHG emissions, and leveraging of digital tools to support climate mitigation and adaptation initiatives, ensuring that digital skills development aligns with broader sustainability goals.	US\$2 million
3.2: Nurturing employment-ready digital skills	US\$3 million
Adaptation and mitigation Efforts undertaken under the program to increase the base of trained digital specialists in the country should assist in ensuring the installation and maintenance of energy-efficient ICT equipment in schools and offices.	
Component 4: Program management and capacity building	US\$7 million

⁷³ There are currently no national energy efficiency performance standards or energy labeling systems in Zambia.

<https://rise.esmap.org/country/zambia>

⁷⁴ Approximately 90 percent of all civil records in Zambia still exist and are stored in paper-based format making them vulnerable and recovery almost impossible should any adverse weather events such as flooding inundate local public government offices, where paper records are kept.



Adaptation and mitigation

Activities under this component include developing practical tools and providing trainings for beneficiaries to enhance and accelerate program implementation, such as sample TOR for procurement (including green procurement principles, for example, factoring resilience of infrastructure and energy efficiency requirements), and model ESS instruments.

TA will be provided to conduct site-specific environmental assessments/plans, notably for large-scale network deployment under Subcomponents 1.1 and 1.2 to identify climate risk mitigation measures to be implemented.



ANNEX 2: Gender

- DZAP conforms to the World Bank's Gender Strategy, 2024–2030,⁷⁵ specifically its Pillar 2 that aims to expand and enable economic opportunities for women.** It adopts an integrated approach toward advancing women's economic empowerment by targeting systemic challenges and gender-specific barriers. For example, women face a gender gap in both internet access and usage, as they not only have fewer opportunities to connect but also tend to use the internet less than their male counterparts. Data from GSMA reveal that, in Sub-Saharan Africa,⁷⁶ women are 37 percent less likely than men to use mobile internet. Additionally, the significance of DPI in service delivery presents a specific challenge for women, creating gender disparities in ID and access to services such as opening a bank account or receiving social assistance payments. In Zambia, 93.6 percent of women and 94.2 percent of men possess IDs.⁷⁷ Achieving meaningful participation of women in these services requires ID verification and the possession of a bank or mobile money account. Women typically encounter more pronounced obstacles in accessing enhanced employment opportunities, primarily stemming from significant gender gaps in acquiring digital skills, among other contributing factors. Additionally, while gender-disaggregated data on digital literacy are limited at the national level, reports show that the region still displays low gender parity scores for digital skills.⁷⁸ Several studies⁷⁹ indicate that the internet, digital platforms, mobiles phones, and DFS provide opportunities to advance and can contribute to reducing the gender divide by enabling women to earn supplementary income, broaden employment options, and access knowledge and general information.
- DZAP seeks to encourage the promotion of economic opportunities for women and foster gender equality in the digital landscape.** Hence, it will be leveraged to narrow identified gender gaps in multiple ways. Through initiatives aiming to provide additional connectivity support and digital skills training for women and female-headed households, the project will encourage women to participate in the digital economy and help them gain employment opportunities. Inclusive enrollment strategies for ID, including the simplification of application and registration procedures, awareness campaigns about the importance of ID, and support for ID enrollment campaigns targeting women will contribute to an increased number of women and girls possessing IDs. Moreover, to encourage productive digital usage, the project will support customized ICT and online safety training dedicated to women, considering any constraints they may encounter.
- Given the project's strategic focus and targeted interventions to address the gender gaps identified, the project's M&E include specific indicators to capture the impact of these interventions and a reduction in the observed gender gaps.** These measures include 11 specific indicators tracking women's access to and productive usage of the internet, including an indicator tracking the number of women who have received a digital ID. The collection of gender-disaggregated data will be key to ensuring DPI is inclusive. Further, the Results Framework includes monitoring of outcomes for female participants in digital skills training. The project's gender analysis results are outlined in table 3.1. It highlights several gender gaps and incorporates specific actions into the project design that align with national gender policies and ICT gender-specific targets.

⁷⁵ World Bank. 2024. *World Bank Group Gender Strategy 2024-2030: Accelerate Gender Equality to Reduce Poverty on a Livable Planet*.

⁷⁶ GSMA. 2021. *The Mobile Gender Gap Report*. <https://www.gsma.com/r/wp-content/uploads/2021/06/The-Mobile-Gender-Gap-Report-2021.pdf>.

⁷⁷ World Bank. 2021. "ID4D Global Dataset."

<https://documents1.worldbank.org/curated/en/099705012232226786/pdf/P176341132c1ef0b21adf11abad304425ef.pdf>.

⁷⁸ ITU. 2022. *Measuring Digital Development Facts and Figures*.

⁷⁹ OECD (Organisation for Economic Co-operation and Development). 2018. *Bridging the Digital Gender Divide*.

<https://www.oecd.org/digital/bridging-the-digital-gender-divide.pdf>.



Table 3.1. Summary Table

Analysis (identified gaps)	Actions (Proposed intervention to address gaps)	Indicators (measures of progress in closing gaps)
Women and girls face a gender gap in both internet access and usage: They experience greater obstacles in connecting to the internet, especially in rural areas, where their access is notably lower than that of their urban and male counterparts. Consequently, women are 15 percent less likely than men to use mobile internet in Sub-Saharan Africa.⁸⁰ In Zambia, for every 8.72 women using the internet, there are an estimated 10 men.⁸¹ This means that 47 percent of internet users are female. Why it matters: Access to the internet empowers women by providing opportunities to utilize financial services, discover job prospects, start businesses, and access educational and health information. It also fosters social networks, encourages participation in community discussions, and promotes engagement in civic activities, ultimately enhancing their social and political inclusion⁸². Root causes of the gender gap: Women face more significant barriers to accessing broadband. These include, among others, lower income levels, cultural norms that may restrict their use of technology, and lower levels of digital literacy.⁸³	Components 1 and 3 of the project aim to bridge the gap in access to internet by empowering women and promoting equitable access to the benefits of the digital economy through the initiatives below. • Infrastructure development: Expanding broadband and mobile networks to rural and underserved areas to increase coverage. Provide targeted support to help female-headed households access broadband services through subsidies, coupons, or vouchers to offset the cost of connectivity and training. • Public access points: Establishing internet access points where women already gather, design specific areas for them to interact with female trainers and staff. Digital literacy programs that consider cultural norms: Offering training to enhance basic and advanced digital skills among women and girls through tailored training that will include modules on online security. To boost women's participation, the project will engage female trainers and utilize devices and technology specifically designed to improve access for women in rural areas. To tackle social norms that may hinder women from attending training, the project will also hire Digital Ambassadors who will customize training sessions to meet the needs and schedule of women. These Digital Ambassadors will offer flexible	People using broadband internet (additional) • <i>Baseline (Dec 2022): 0</i> • <i>Target (Oct 203): 7.4 million</i> <i>Of which, female: 3.7 million</i> • <i>New users: 1.7 million</i> • <i>Enhanced users: 1 million</i> • <i>Inferred users: 1 million</i>

⁸⁰ https://www.gsma.com/r/gender-gap/?utm_source=website&utm_medium=blog&utm_campaign=gender-gap-2023

⁸¹ <https://digitalgendergaps.org/>, using the Facebook Gender Gap Index.

⁸² <https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/blog/closing-the-mobile-gender-gap-in-sub-saharan-africa-a-social-and-commercial-opportunity/#:~:text=Connecting%20all%20women%20to%20mobile,services%20and%20life%2Denhancing%20opportunities>.

⁸³ <https://documents1.worldbank.org/curated/en/099052923081020509/pdf/P177348030d77b00f09db70d49e03be1574.pdf>



Analysis (identified gaps)	Actions (Proposed intervention to address gaps)	Indicators (measures of progress in closing gaps)
	hours and schedule sessions outside of peak household responsibilities, allowing women to participate without disrupting their daily routines.	
Women are slightly less likely than men to own a digital ID which reduces their ability to take advantage of digital services: In Zambia 93.6 percent of females and 94.2 percent of males possess IDs. Disaggregated data is not available for digital ID possession. But while the difference in ID possession is not significant, given the existing gap in the use of digital tools, women may also be less likely to own digital IDs. Why it matters: With a digital ID, women gain the ability to formally register businesses, secure land ownership, obtain formal employment, open bank accounts, register SIM cards for mobile phone use, access DFS, and receive government financial support more easily and legally.⁸⁴ Root causes of the gender gap: (a) Cultural and social barriers: Traditional gender roles and cultural norms restrict women's mobility and access to necessary documentation processes; (b) Economic constraints: Many women face financial barriers that prevent them from obtaining IDs, such as fees associated with registration and travel costs to registration centers; (c) Lack of awareness: Women may not be aware of the importance of digital IDs or the processes to obtain them; (d) Education and literacy levels: Lower levels of education and literacy among women can make navigating the digital ID application process more challenging.⁸⁵	Under Component 2, the project contributes to an increased number of women and girls possessing digital IDs by: - Developing tailored awareness campaigns (that is, word of mouth, radio, SMS, print media, and social media) about the importance of ID for women and the benefits for their households; - Providing support to ID enrollment campaigns targeting women and further guiding them in the process; - Designing inclusive enrollment strategies for ID, including the simplification of application and registration procedures, with enrollment facilitated by agents to help address mobility, financial, and literacy constraints; and - Ensuring enrollment centers are sufficiently staffed, including female personnel, to reduce processing times and provide ample seating, restrooms, shaded waiting areas, and services in local languages where applicable.	People with digitally verifiable identification Baseline (Dec 2022): 1.5m <i>Of which female: 0 (not available)</i> Target (Oct 2030): 4 million <i>Of which female: 2 million (50%)</i>
Women frequently find themselves with less proficiency in digital skills compared to men, resulting in limited access to improved job prospects and economic opportunities: While gender-disaggregated data on digital literacy are limited at the national level, reports show that the region still displays low gender parity scores for digital skills.⁸⁶	Under Component 3, the project encourages productive digital usage by women and girls and economic empowerment via digital skills training by Providing customized and certified digital skills and online safety training dedicated to women,	Graduates of advanced digital skills training programs Baseline (Dec 2023): 0 Target (Oct 2030): 1,000 Of which female: 500 (50%)

⁸⁴ <https://id4d.worldbank.org/global-dataset#highlights>.

⁸⁵ <https://documents1.worldbank.org/curated/en/953621531854471275/Global-ID-Coverage-Barriers-and-Use-by-the-Numbers-Insights-from-the-ID4D-Findex-Survey.pdf>

⁸⁶ ITU. 2022. *Measuring Digital Development Facts and Figures*.



Analysis (identified gaps)	Actions (Proposed intervention to address gaps)	Indicators (measures of progress in closing gaps)
As an example, the proportion of females and males who made or received a digital payment during the survey year (2021) in Zambia⁸⁷ is as follows: • Males, 48 percent • Females, 44 percent Due to the absence of statistics on advanced skills, the existing data primarily focus on basic skills. However, these figures suggest a potential ongoing gender disparity that may extend into more complex digital competencies. Why it matters: Without digital skills, women face significant barriers in advancing themselves, including earning additional income, getting new employment opportunities, acquiring knowledge and accessing information.⁸⁸ Root causes of the gender gap: Among other factors, lack of education as well as inherent biases and sociocultural norms curtail women and girls' ability to benefit from the opportunities offered by the digital transformation.	taking into consideration any constraints (physical mobility, time) they may encounter; • Developing advanced certified training courses, including for women, and tailored to the private sector needs in HEIs and tech hubs; and • Including supplemental assessments in digital skills training. The certification process for both basic and advanced training will feature a well-structured curriculum that encompasses practical exams, real-world problem-solving exercises, and continuous evaluations. This approach guarantees that candidates demonstrate proficiency in essential digital tools pertinent to their specific training pathway. The accreditation will adhere to industry standards, and instruction will be provided by highly qualified experts with a proven track record. • To address social norms and biases toward women, the project will engage women by featuring female role models, such as those currently working in technology. It will also implement mentorship programs and community events with women leaders in tech to foster self-efficacy. Additionally, the project will involve the community, including men, to build trust and mitigate potential exclusion.	

⁸⁷ World Bank. 2024. "Findex Survey."

⁸⁸ OECD. 2018. *Bridging the Digital Gender Divide*. <https://www.oecd.org/digital/bridging-the-digital-gender-divide.pdf>.